1 Structure and Bonding

Goals for Chapter 1

1 Review concepts from general chemistry that are essential for success in organic chemistry, such as the electronic structure of the atom, Lewis structures and the octet rule, types of bonding, electronegativity, and formal charges.

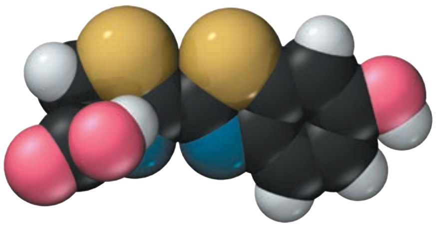
2 Predict patterns of covalent and ionic bonding involving C, H, O, N, and the halogens.

3 Identify resonance-stabilized structures and compare the relative importance of their resonance forms.

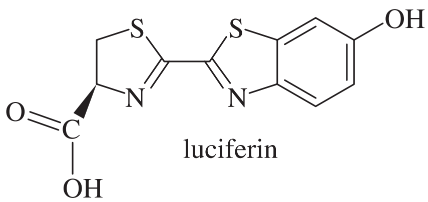
4 Draw and interpret the types of structural formulas commonly used in organic chemistry, including condensed structural formulas and line–angle formulas.

5 Predict the hybridization and geometry of organic molecules based on their bonding.

6 Identify isomers and explain the differences between them.



luciferin



luciferin

◄ **Luciferin** is the light-emitting compound found in many firefly (Lampyridae) species. Luciferin reacts with atmospheric oxygen, under the control of an enzyme, to emit the yellow light that fireflies use to attract mates or prey.

1-1 The Origins of Organic Chemistry

The modern definition of organic chemistry is *the chemistry of carbon compounds*. What is so special about carbon that a whole branch of chemistry is devoted to its compounds? Unlike most other elements, carbon forms strong bonds to other carbon atoms and to a wide variety of other elements. Chains and rings of carbon atoms can be built up to form an endless variety of molecules. This diversity of carbon compounds provides the basis for life on Earth. Living creatures are composed largely of complex organic compounds that serve structural, chemical, or genetic functions.

The term **organic** literally means “derived from living organisms.” Originally, the science of organic chemistry was the study of compounds extracted from living organisms and their natural products. Compounds such as sugar, urea, starch, waxes, and plant oils were considered “organic,” and people accepted **vitalism,** the belief that natural products needed a “vital force” to create them. Organic chemistry, then, was the study of compounds having the vital force. Inorganic chemistry was the study of gases, rocks, and minerals, and the compounds that could be made from them.

Application: Drug Research

One of the reasons chemists synthesize derivatives of complex organic compounds such as morphine (shown below) is to discover new drugs that retain the good properties (potent pain-relieving) but not the bad properties (highly addictive).

In the 19th century, experiments showed that organic compounds could be synthesized from inorganic compounds. In 1828, the German chemist Friedrich Wöhler converted ammonium cyanate, made from ammonia and cyanic acid, to urea simply by heating it in the absence of oxygen.

$$\underset{\substack{\text{ammonium cyanate}\\\text{(inorganic)}}}{NH_4^+ \; {}^-OCN} \xrightarrow{\text{heat}} \underset{\substack{\text{urea}\\\text{(organic)}}}{H_2N{-}\overset{\overset{\displaystyle O}{\|}}{C}{-}NH_2}$$

Urea had always come from living organisms and was presumed to contain the vital force, yet ammonium cyanate is inorganic and thus lacks the vital force. Some chemists claimed that a trace of vital force from Wöhler's hands must have contaminated the reaction, but most recognized the possibility of synthesizing organic compounds from inorganics. Many other syntheses were carried out, and the vital force theory was eventually discarded.

Because vitalism was disproved in the early 19th century, you'd think it would be extinct by now. And you'd be wrong! Vitalism lives on today in the minds of those who believe that "natural" (plant-derived) vitamins, flavor compounds, etc., are somehow different and more healthful than the identical "artificial" (synthesized) compounds.

As chemists, we know that plant-derived compounds and the synthesized compounds are identical. Assuming they are pure, the only way to tell them apart is through ^{14}C dating: Compounds synthesized from petrochemicals have a lower content of radioactive ^{14}C and appear old because their ^{14}C has decayed over time. Plant-derived compounds are recently synthesized from CO_2 in the air. They have a higher content of radioactive ^{14}C. Some large chemical suppliers provide isotope-ratio analyses to show that their "naturals" have high ^{14}C content and are plant-derived. Such a sophisticated analysis lends a high-tech flavor to this 21st-century form of vitalism.

Even though organic compounds do not need a vital force, they are still distinguished from inorganic compounds. The distinctive feature of organic compounds is that they *all* contain one or more carbon atoms. Still, not all carbon compounds are organic;

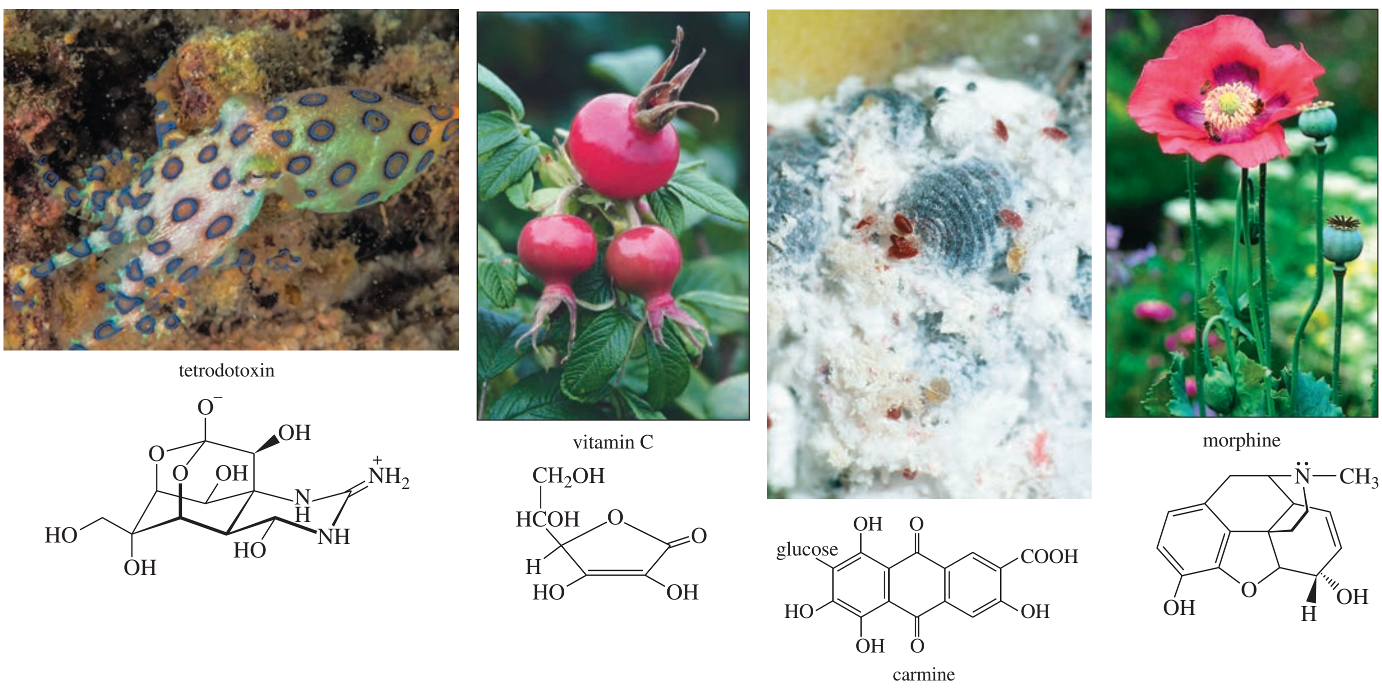


FIGURE 1-1 (a) Venom of the blue-ringed octopus contains tetrodotoxin, which causes paralysis resulting in death. (b) Rose hips contain vitamin C, a radical inhibitor (Chapter 4) that prevents scurvy. (c) The prickly pear cactus is host to cochineal insects, used to prepare the red dye carmine (Chapter 15). (d) Opium poppies contain morphine, an addictive, pain-relieving alkaloid (Chapter 19).

substances such as diamond, graphite, carbon dioxide, ammonium cyanate, and sodium carbonate are derived from minerals and have typical inorganic properties. Most of the millions of carbon compounds are classified as organic, however.

We humans are composed largely of organic molecules, and we are nourished by the organic compounds in our food. The proteins in our skin, the lipids in our cell membranes, the glycogen in our livers, and the DNA in the nuclei of our cells are all organic compounds. Our bodies are also regulated and defended by complex organic compounds.

Chemists have learned to synthesize or simulate many of these complex molecules. The synthetic products serve as drugs, medicines, plastics, pesticides, paints, and fibers. Many of the most important advances in medicine are actually advances in organic chemistry. New synthetic drugs are developed to combat disease, and new polymers are molded to replace failing organs. Organic chemistry has gone full circle. It began as the study of compounds derived from "organs," and now it gives us the drugs and materials we need to save or replace those organs.

The AbioCor® self-contained artificial heart, which is used to sustain patients who are waiting for a heart transplant. The outer shell is polycarbonate, and the valves and inner bladder are polyurethane. Both of these durable substances are synthetic organic compounds.

1-2 Principles of Atomic Structure

Before we begin our study of organic chemistry, we must review some basic principles. These concepts of atomic and molecular structure are crucial to your understanding of the structure and bonding of organic compounds.

1-2A Structure of the Atom

Atoms are made up of protons, neutrons, and electrons. Protons are positively charged and are found together with (uncharged) neutrons in the nucleus. Electrons, which have a negative charge that is equal in magnitude to the positive charge on the proton, occupy the space surrounding the nucleus (Figure 1-2). Protons and neutrons have similar masses, about 1800 times the mass of an electron. Almost all the atom's mass is in the nucleus, but it is the electrons that take part in chemical bonding and reactions.

Each element is distinguished by the number of protons in the nucleus (the atomic number). The number of neutrons is usually similar to the number of protons, although the number of neutrons may vary. Atoms with the same number of protons but different numbers of neutrons are called **isotopes.** For example, the most common kind of carbon atom has six protons and six neutrons in its nucleus. Its mass number (the sum of the protons and neutrons) is 12, and we write its symbol as ^{12}C. About 1% of carbon atoms have seven neutrons; the mass number is 13, written ^{13}C. A very small fraction of carbon atoms have eight neutrons and a mass number of 14. The ^{14}C isotope is radioactive, with a half-life (the time it takes for half of the nuclei to decay) of 5730 years. The predictable decay of ^{14}C is used to determine the age of organic materials up to about 50,000 years old.

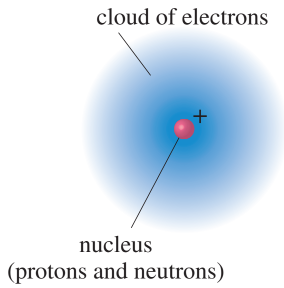
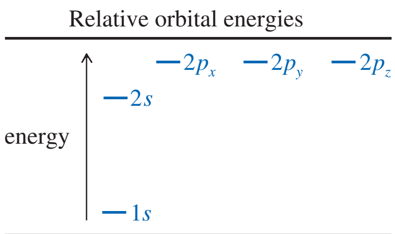


FIGURE 1-2 Basic atomic structure. An atom has a dense, positively charged nucleus surrounded by a cloud of electrons.

1-2B Electron Shells and Orbitals

An element's chemical properties are determined by the number of protons in the nucleus and the corresponding number of electrons around the nucleus. The electrons form bonds and determine the structure of the resulting molecules. Because they are small and light, electrons show properties of both particles and waves; in many ways, the electrons in atoms and molecules behave more like waves than like particles.

Electrons that are bound to nuclei are found in **orbitals.** Orbitals are mathematical descriptions that chemists use to explain and predict the properties of atoms and molecules. The *Heisenberg uncertainty principle* states that we can never determine exactly where the electron is; nevertheless, we can determine the **electron density,** the probability of finding the electron in a particular part of the orbital. An orbital, then, is an allowed energy state for an electron, with an associated probability function that defines the distribution of electron density in space.



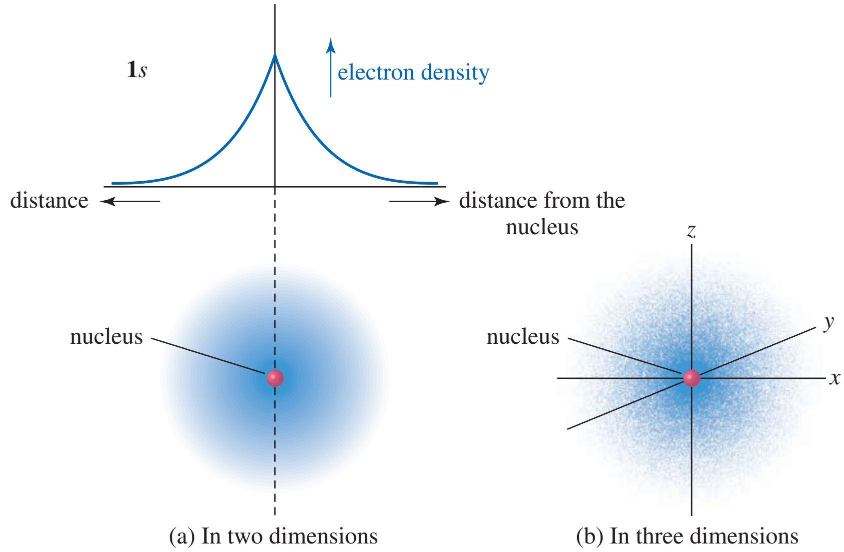


FIGURE 1-3 Graph and diagram of the 1*s* atomic orbital. The electron density is highest at the nucleus and drops off exponentially with increasing distance from the nucleus in any direction.

Atomic orbitals are grouped into different "shells" at different distances from the nucleus. Each shell is identified by a principal quantum number n, with $n = 1$ for the lowest-energy shell closest to the nucleus. As n increases, the shells are farther from the nucleus, are higher in energy, and can hold more electrons. Most of the common elements in organic compounds are found in the first two rows of the periodic table, indicating that their electrons are found in the first two electron shells. The first shell ($n = 1$) can hold two electrons, and the second shell ($n = 2$) can hold eight.

The first electron shell contains just the 1*s* orbital. All *s* orbitals are spherically symmetrical, meaning that they are nondirectional. The electron density is only a function of the distance from the nucleus. The electron density of the 1*s* orbital is graphed in Figure 1-3. Notice how the electron density is highest *at* the nucleus and falls off exponentially with increasing distance from the nucleus. The 1*s* orbital might be imagined as a cotton boll, with the cottonseed at the middle representing the nucleus. The density of the cotton is highest nearest the seed, and it becomes less dense at greater distances from this "nucleus."

The second electron shell consists of the 2*s* and 2*p* orbitals. The 2*s* orbital is spherically symmetrical like the 1*s* orbital, but its electron density is not a simple exponential function. The 2*s* orbital has a smaller amount of electron density close to the nucleus. Most of the electron density is farther away, beyond a region of zero electron density called a **node.** Because most of the 2*s* electron density is farther from the nucleus than that of the 1*s*, the 2*s* orbital is higher in energy. Figure 1-4 shows a graph of the 2*s* orbital.

In addition to the 2*s* orbital, the second shell also contains three 2*p* atomic orbitals, one oriented in each of the three spatial directions. These orbitals are called the $2p_x$, the $2p_y$, and the $2p_z$, according to their direction along the x, y, or z axis. The 2*p* orbitals are slightly higher in energy than the 2*s*, because the average location of the electron in a 2*p* orbital is farther from the nucleus. Each *p* orbital consists of two lobes, one on either side of the nucleus, with a **nodal plane** at the nucleus. The nodal plane is a flat (planar) region of space, including the nucleus, with zero electron density. The three 2*p* orbitals differ only in their spatial orientation, so they have identical energies. Orbitals with identical energies are called **degenerate orbitals.** Figure 1-5 shows the shapes of the three degenerate 2*p* atomic orbitals.

The *Pauli exclusion principle* tells us that each orbital can hold a maximum of two electrons, provided that their spins are paired. The first shell (one 1*s* orbital) can accommodate two electrons. The second shell (one 2*s* orbital and three 2*p* orbitals) can accommodate eight electrons, and the third shell (one 3*s* orbital, three 3*p* orbitals, and five 3*d* orbitals) can accommodate 18 electrons.

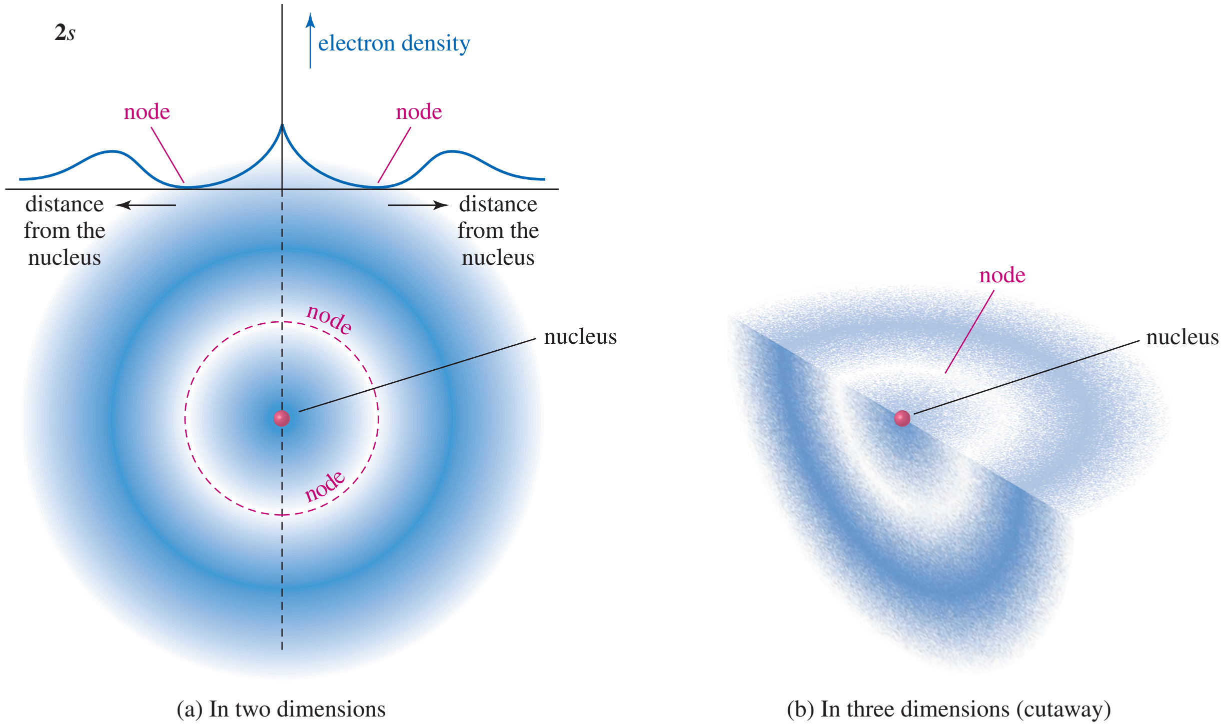


FIGURE 1-4 Graph and diagram of the 2*s* atomic orbital. The 2*s* orbital has a small region of high electron density close to the nucleus, but most of the electron density is farther from the nucleus, beyond a node, or region of zero electron density.

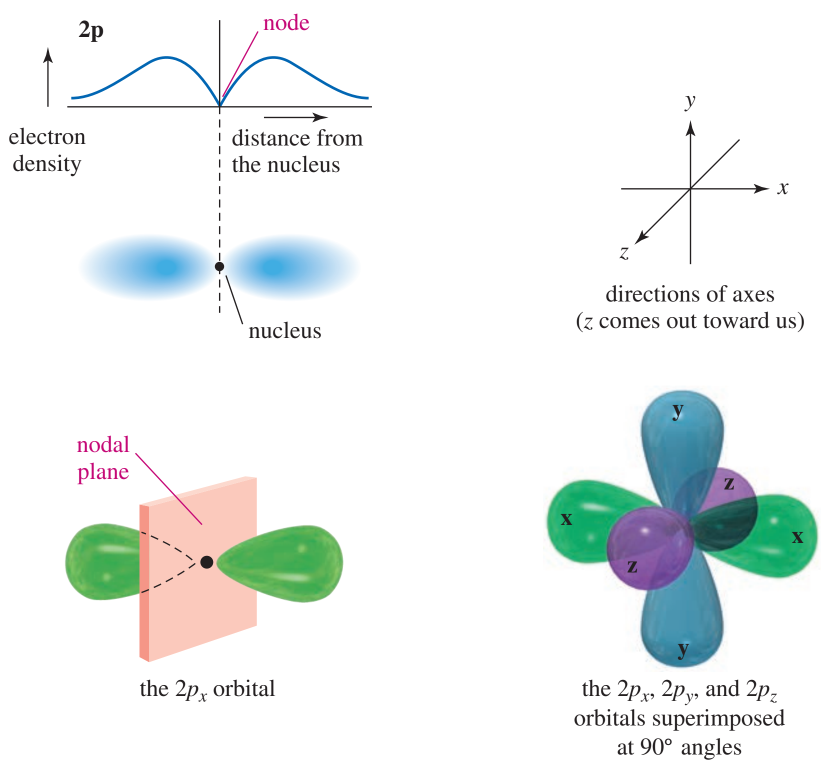


FIGURE 1-5 The 2*p* orbitals. Three 2*p* orbitals are oriented at right angles to each other. Each is labeled according to its orientation along the *x*, *y*, or *z* axis.

Application: Drugs

Lithium carbonate, a salt of lithium, is a mood-stabilizing agent used to treat the psychiatric disorder known as mania. Mania is characterized by behaviors such as elated mood, feelings of greatness, racing thoughts, and an inability to sleep. We don't know how lithium carbonate helps to stabilize these patients' moods.

Application: Ibuprofen

Ibuprofen (chemical formula: $C_{13}H_{18}O_2$) is a common nonmetallic drug sold at the pharmacy. It belongs to the class of nonsteroidal anti-inflammatory drugs (NSAIDs) that are used for treating pain, fever, and inflammation. It is administered to relieve headache, dental pain, menstrual cramps, muscle aches, and arthritis.

1-2C Electronic Configurations of Atoms

Aufbau means "building up" in German, and the *aufbau principle* tells us how to build up the electronic configuration of an atom's ground (most stable) state. Starting with the lowest-energy orbital, we fill the orbitals in order until we have added the proper number of electrons. Table 1-1 shows the ground-state electronic configurations of the elements in the first two rows of the periodic table.

Two additional concepts are illustrated in Table 1-1. The **valence electrons** are those electrons in the outermost shell. Carbon has four valence electrons, nitrogen has five, and oxygen has six. Helium has a filled first shell with two valence electrons, and neon has a filled second shell with eight valence electrons (ten electrons total). In general (for the representative elements), the column or group number of the periodic table corresponds to the number of valence electrons (Figure 1-6). Hydrogen and lithium have one valence electron, and they are both in the first column (group 1A) of the periodic table. Carbon has four valence electrons, and it is in group 4A of the periodic table.

Notice in Table 1-1 that carbon's third and fourth valence electrons are not paired; they occupy separate orbitals. Although the Pauli exclusion principle says that two electrons can occupy the same orbital, the electrons repel each other, and pairing requires additional energy. **Hund's rule** states that when there are two or more orbitals of the same energy, electrons go into *different* orbitals rather than pair up in the same orbital. The first $2p$ electron (boron) goes into one $2p$ orbital, the second $2p$ electron (carbon) goes into a different orbital, and the third $2p$ electron (nitrogen) occupies the last $2p$ orbital. The fourth, fifth, and sixth $2p$ electrons must pair up with the first three electrons.

TABLE 1-1 Electronic Configurations of the Elements of the First and Second Rows

Element	Configuration	Valence Electrons
H	$1s^1$	1
He	$1s^2$	2
Li	$1s^2 2s^1$	1
Be	$1s^2 2s^2$	2
B	$1s^2 2s^2 2p_x^1$	3
C	$1s^2 2s^2 2p_x^1 2p_y^1$	4
N	$1s^2 2s^2 2p_x^1 2p_y^1 2p_z^1$	5
O	$1s^2 2s^2 2p_x^2 2p_y^1 2p_z^1$	6
F	$1s^2 2s^2 2p_x^2\, 2p_y^2\, 2p_z^1$	7
Ne	$1s^2 2s^2 2p_x^2\, 2p_y^2\, 2p_z^2$	8

Partial periodic table

1A							noble gases 8A
H	2A	3A	4A	5A	6A	7A	He
Li	Be	B	C	N	O	F	Ne
Na	Mg	Al	Si	P	S	Cl	Ar

FIGURE 1-6 First three rows of the periodic table. The organization of the periodic table results from the filling of atomic orbitals in order of increasing energy. For these representative elements, the number of the column corresponds to the number of valence electrons.

PROBLEM 1-1

(a) Nitrogen has relatively stable isotopes (half-life greater than 1 second) of mass numbers 13, 14, 15, 16, and 17. (All except ^{14}N and ^{15}N are radioactive.) Calculate how many protons and neutrons are in each of these isotopes of nitrogen.
(b) Write the electronic configurations of the third-row elements shown in the partial periodic table in Figure 1-6.

1-3 Bond Formation: The Octet Rule

PROBLEM-SOLVING HINT

When we speak of a molecule having "all octets satisfied," we mean that all the second-row elements have octets. Hydrogen atoms have just two electrons (the He configuration) in their filled valence shell.

In 1915, G. N. Lewis proposed several new theories describing how atoms bond together to form molecules. One of these theories states that a filled shell of electrons is especially stable, and *atoms transfer or share electrons in such a way as to attain a filled shell of electrons*. A filled shell of electrons is simply the electron configuration of a noble gas, such as He, Ne, or Ar. This principle has come to be called the **octet rule** because a filled shell implies eight valence electrons for the elements in the second row of the periodic table. Elements in the third and higher rows (such as Al, Si, P, S, Cl, and above) can have an "expanded octet" of more than eight electrons because they have low-lying *d* orbitals available.

1-3A Ionic Bonding

There are two ways that atoms can interact to attain noble-gas configurations. Sometimes atoms attain noble-gas configurations by transferring electrons from one atom to another. For example, lithium has one electron more than the helium configuration, and fluorine has one electron less than the neon configuration. Lithium easily loses its valence electron, and fluorine easily gains one:

$$\underset{\text{electron transfer}}{\mathrm{Li}\cdot\ \cdot\ddot{\underset{..}{\mathrm{F}}}:} \longrightarrow \underset{\text{He configuration}}{\mathrm{Li}^+} + \underset{\text{Ne configuration}}{:\ddot{\underset{..}{\mathrm{F}}}:^-} \longrightarrow \underset{\text{ionic bond}}{\mathrm{Li}^+\,:\ddot{\underset{..}{\mathrm{F}}}:^-}$$

A transfer of one electron gives each of these two elements a noble-gas configuration. The resulting ions have opposite charges, and they attract each other to form an **ionic bond.** Ionic bonding usually results in the formation of a large crystal lattice rather than individual molecules. Ionic bonding is common in inorganic compounds but relatively uncommon in organic compounds.

1-3B Covalent Bonding

Covalent bonding, in which electrons are shared rather than transferred, is the most common type of bonding in organic compounds. Hydrogen, for example, needs a second electron to achieve the noble-gas configuration of helium. If two hydrogen atoms come together and form a bond, they "share" their two electrons, and each atom has two electrons in its valence shell.

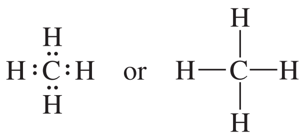
$$\mathrm{H}\cdot \ + \ \mathrm{H}\cdot \longrightarrow \underset{\text{covalent bond}}{\mathrm{H}:\mathrm{H}} \qquad \text{each H shares two electrons (He configuration)}$$

We will study covalent bonding in more detail later in this chapter.

1-4 Lewis Structures

One way to symbolize the bonding in a covalent molecule is to use **Lewis structures.** In a Lewis structure, each valence electron is symbolized by a dot. A bonding pair of electrons is symbolized by a pair of dots or by a dash (—). We try to arrange all the atoms so that they have their appropriate noble-gas configurations: two electrons for hydrogen, and octets for the second-row elements.

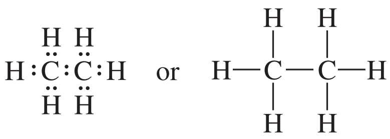
Consider the Lewis structure of methane (CH_4).



methane

Carbon contributes four valence electrons, and each hydrogen contributes one, to give a total of eight electrons. All eight electrons surround carbon to give it an octet, and each hydrogen atom shares two of the electrons with the carbon atom.

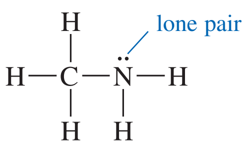
The Lewis structure for ethane (C_2H_6) is more complex.



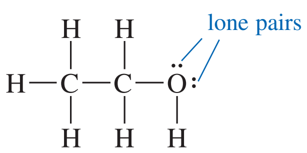
ethane

Once again, we have computed the total number of valence electrons (14) and distributed them so that each carbon atom is surrounded by 8 and each hydrogen by 2. The only possible structure for ethane is the one shown, with the two carbon atoms sharing a pair of electrons and each hydrogen atom sharing a pair with one of the carbons. The ethane structure shows the most important characteristic of carbon—its ability to form strong carbon–carbon bonds.

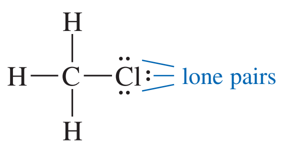
Nonbonding electrons are valence-shell electrons that are *not* shared between two atoms. A pair of nonbonding electrons is often called a **lone pair.** Oxygen atoms, nitrogen atoms, and the halogens (F, Cl, Br, I) usually have nonbonding electrons in their stable compounds. These lone pairs of nonbonding electrons often serve as reactive sites in their parent compounds. The following Lewis structures show one lone pair of electrons on the nitrogen atom of methylamine and two lone pairs on the oxygen atom of ethanol. Halogen atoms usually have three lone pairs, as shown in the structure of chloromethane.



methylamine



ethanol



chloromethane

A correct Lewis structure should show any lone pairs. Organic chemists often draw structures that omit most or all of the lone pairs. These are not true Lewis structures because you must imagine the correct number of nonbonding electrons.

PROBLEM-SOLVING HINT

Lewis structures are the way we write organic chemistry. Learning how to draw them quickly and correctly will help you throughout this course.

PROBLEM 1-2

Draw Lewis structures for the following compounds.

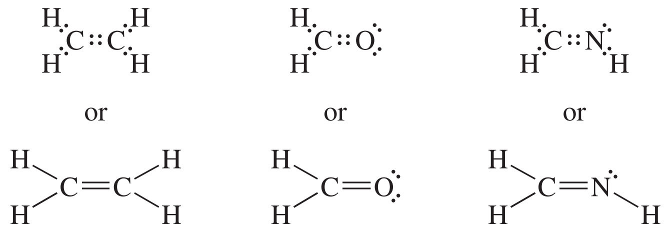
(a) ammonia, NH_3
(b) water, H_2O
(c) hydronium ion, H_3O^+
(d) propane, C_3H_8
(e) dimethylamine, CH_3NHCH_3
(f) diethyl ether, $CH_3CH_2OCH_2CH_3$
(g) 1-chloropropane, $CH_3CH_2CH_2Cl$
(h) propan-2-ol, $CH_3CH(OH)CH_3$
(i) borane, BH_3
(j) boron trifluoride, BF_3

Explain what is unusual about the bonding in the compounds in parts (i) and (j).

1-5 Multiple Bonding

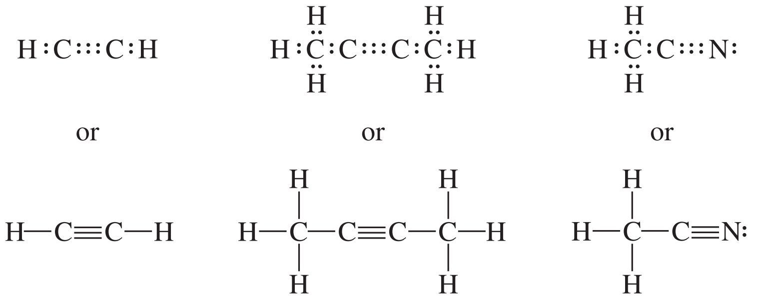
In drawing Lewis structures in Section 1-4, we placed just one pair of electrons between any two atoms. The sharing of one pair between two atoms is called a **single bond.** Many molecules have adjacent atoms sharing two or even three electron pairs. The sharing of two pairs is called a **double bond,** and the sharing of three pairs is called a **triple bond.**

Ethylene (C_2H_4) is an organic compound with a double bond. When we draw a Lewis structure for ethylene, the only way to show both carbon atoms with octets is to draw them sharing two pairs of electrons. The following examples show organic compounds with double bonds. In each case, two atoms share four electrons (two pairs) to give them octets. A double dash (=) symbolizes a double bond.



ethylene formaldehyde formaldimine

Acetylene (C_2H_2) has a triple bond. Its Lewis structure shows three pairs of electrons between the carbon atoms to give them octets. The following examples show organic compounds with triple bonds. A triple dash (≡) symbolizes a triple bond.



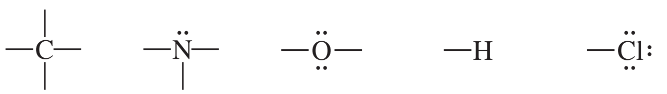
acetylene dimethylacetylene acetonitrile

All these Lewis structures show that carbon normally forms four bonds in neutral organic compounds. Nitrogen generally forms three bonds, and oxygen usually forms two. Hydrogen and the halogens usually form only one bond. The number of bonds an atom usually forms is called its **valence.** Carbon is tetravalent, nitrogen is trivalent, oxygen is divalent, and hydrogen and the halogens are monovalent. By remembering the usual number of bonds for these common elements, we can write organic structures more easily. If we draw a structure with each atom having its usual number of bonds, a correct Lewis structure usually results.

Application: FYI

Acetylene is a high-energy gaseous hydrocarbon that is explosive at high pressures. Combined with oxygen, acetylene burns with such a hot flame that it melts steel. Acetylene is commonly used in welding and cutting torches that work anywhere, even underwater. In gas cylinders, acetylene is dissolved in acetone to keep it from getting too concentrated and exploding.

SUMMARY Common Bonding Patterns (Uncharged)



	carbon	nitrogen	oxygen	hydrogen	halogens
valence:	4	3	2	1	1
lone pairs:	0	1	2	0	3

PROBLEM-SOLVING HINT

These "usual numbers of bonds" might be single bonds, or they might be combined into double and triple bonds. For example, three bonds to nitrogen might be three single bonds, one single bond and one double bond, or one triple bond (:N≡N:). In working problems, consider all possibilities.

PROBLEM 1-3

Write Lewis structures for the following molecular formulas.

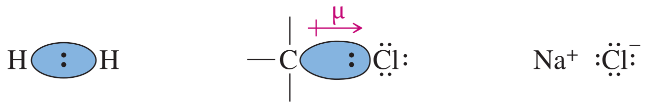
(a) N_2 **(b)** HCN **(c)** HONO
(d) CO_2 **(e)** CH_3CHNH **(f)** HCO_2H
(g) C_2H_3Cl **(h)** HNNH **(i)** C_3H_6 (one double bond)
(j) C_3H_4 (two double bonds) **(k)** C_3H_4 (one triple bond)

PROBLEM 1-4

Circle any lone pairs (pairs of nonbonding electrons) in the structures you drew for Problem 1-3.

1-6 Electronegativity and Bond Polarity

A bond with the electrons shared equally between the two atoms is called a **nonpolar covalent bond.** The bond in H_2 and the C—C bond in ethane are nonpolar covalent bonds. In most bonds between two different elements, the bonding electrons are attracted more strongly to one of the two nuclei. An unequally shared pair of bonding electrons is called a **polar covalent bond.**



nonpolar covalent bond | polar covalent bond | ionic bond

When carbon is bonded to chlorine, for example, the bonding electrons are attracted more strongly to the chlorine atom. The carbon atom bears a small partial positive charge, and the chlorine atom bears a partial negative charge. Figure 1-7 shows the polar carbon–chlorine bond in chloromethane. We symbolize the bond polarity using an arrow with its head at the negative end of the polar bond and a plus sign at the positive end. The bond polarity is measured by its **dipole moment** (μ), defined to be the amount of partial charge (δ^+ and δ^-) multiplied by the bond length (d). The symbol δ^+ means "a small amount of positive charge"; δ^- means "a small amount of negative charge." To symbolize a dipole moment, we use a crossed arrow pointing from the + charge toward the – charge.

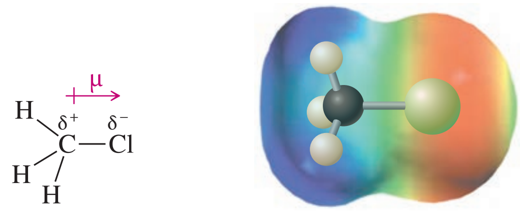
$$\mu = \delta \times d$$
dipole moment

Figure 1-7 also shows an **electrostatic potential map (EPM)** for chloromethane, using color to represent the calculated charge distribution in the molecule. Red shows electron-rich regions. Blue and purple show electron-poor regions. Orange, yellow, and green show intermediate levels of electrostatic potential. In chloromethane, the red region shows the partial negative charge on chlorine, and the blue region shows the partial positive charges on carbon and the hydrogen atoms.

We often use **electronegativities** as a guide in predicting whether a given bond will be polar and the direction of its dipole moment. The Pauling electronegativity scale, most commonly used by organic chemists, is based on bonding properties, and it is useful for predicting the polarity of covalent bonds. Elements with higher electronegativities generally have more attraction for the bonding electrons. Therefore, in a bond between two different atoms, the atom with the higher electronegativity is the negative end of the dipole. Figure 1-8 shows Pauling electronegativities for some of the important elements in organic compounds.

Linus Carl Pauling (1901–1994) was an American chemist. Pauling was a pioneer in the field of quantum chemistry. He introduced the concept of orbital hybridization and established the electronegativity scale of the elements. Pauling is the only person to win two unshared Nobel Prizes, having been awarded the Nobel Prize in Chemistry (1954) and the Nobel Peace Prize (1962).

Notice that the electronegativities increase from left to right across the periodic table. Nitrogen, oxygen, and the halogens are all more electronegative than carbon; sodium, lithium, and magnesium are less electronegative. Hydrogen's electronegativity is similar to that of carbon, so we usually consider C—H bonds to be nonpolar. We will consider the polarity of bonds and molecules in more detail in Section 2-1.



chloromethane chloromethane

FIGURE 1-7 Bond polarity. Chloromethane contains a polar carbon–chlorine bond with a partial negative charge on chlorine and a partial positive charge on carbon. The electrostatic potential map shows a red region (electron-rich) around the partial negative charge and a blue region (electron-poor) around the partial positive charge. Other colors show intermediate values of electrostatic potential.

H 2.2						
Li 1.0	Be 1.6	B 2.0	C 2.5	N 3.0	O 3.4	F 4.0
Na 0.9	Mg 1.3	Al 1.6	Si 1.9	P 2.2	S 2.6	Cl 3.2
K 0.8						Br 3.0
						I 2.7

FIGURE 1-8 The Pauling electronegativities of some of the elements found in organic compounds.

PROBLEM 1-5

Use electronegativities to predict the direction of the dipole moments of the following bonds.

(a) C—Br **(b)** C—I **(c)** C—Si **(d)** B—Cl **(e)** B—F
(f) N—F **(g)** N—Br **(h)** Si—Br **(i)** Si—Cl **(j)** Si—F

1-7 Formal Charges

In polar bonds, the partial charges (δ^+ and δ^-) on the bonded atoms are *real*. **Formal charges** provide a method for keeping track of electrons, but they may or may not correspond to real charges. In most cases, if the Lewis structure shows that an atom has a formal charge, it actually bears at least part of that charge. The concept of formal charge helps us determine which atoms bear most of the charge in a charged molecule; it also helps us to see charged atoms in molecules that are neutral overall.

To calculate formal charges, count how many electrons contribute to the charge of each atom and compare that number with the number of valence electrons in the free, neutral atom (given by the group number in the periodic table on the inside back cover). The electrons that contribute to an atom's charge are

1. *all* its unshared (nonbonding) electrons; plus
2. *half* the (bonding) electrons it shares with other atoms, or one electron of each bonding pair.

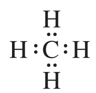
The formal charge of a given atom can be calculated by the formula

$$\text{formal charge (FC)} = [\text{group number}] - [\text{nonbonding electrons}] - \tfrac{1}{2}[\text{shared electrons}]$$

SOLVED PROBLEM 1-1

Compute the formal charge (FC) on each atom in the following structures.

(a) Methane (CH_4)



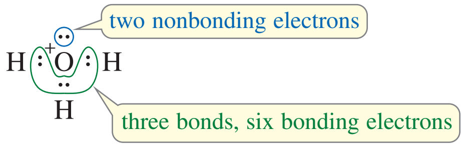
SOLUTION

Each of the hydrogen atoms in methane has one bonding pair of electrons (two shared electrons). Half of two shared electrons is one electron, and one valence electron is what hydrogen needs to be neutral. Hydrogen atoms with one bond are formally neutral: FC = 1 − 0 − 1 = 0.

(continued)

The carbon atom has four bonding pairs of electrons (eight electrons). Half of eight shared electrons is four electrons, and four electrons are what carbon (group 4A) needs to be neutral. Carbon is formally neutral whenever it has four bonds: FC $= 4 - 0 - \frac{1}{2}(8) = 0$.

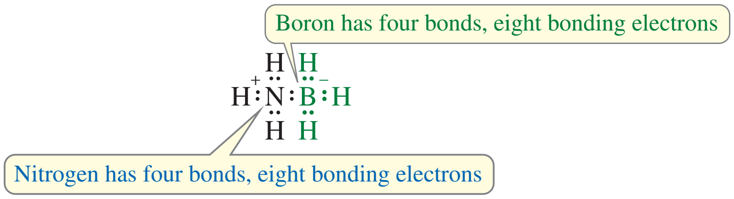
(b) The hydronium ion, H_3O^+



SOLUTION

In drawing the Lewis structure for this ion, we use eight electrons: six from oxygen plus three from the hydrogens, minus one because the ion has a positive charge. Each hydrogen has one bond and is formally neutral. Oxygen is surrounded by an octet, with six bonding electrons and two nonbonding electrons. Half the bonding electrons plus all the nonbonding electrons contribute to its charge: $\frac{6}{2} + 2 = 5$; but oxygen (group 6A) needs six valence electrons to be neutral. Consequently, the oxygen atom has a formal charge of $+1$: FC $= 6 - 2 - \frac{1}{2}(6) = +1$.

(c) $H_3N{-}BH_3$



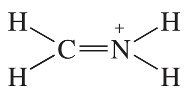
SOLUTION

This is a neutral compound where the individual atoms are formally charged. The Lewis structure shows that both nitrogen and boron have four shared bonding pairs of electrons. Both boron and nitrogen have $\frac{8}{2} = 4$ electrons contributing to their charges. Nitrogen (group 5A) needs five valence electrons to be neutral, so it bears a formal charge of $+1$. Boron (group 3A) needs only three valence electrons to be neutral, so it bears a formal charge of -1.

Nitrogen: FC $= 5 - 0 - \frac{1}{2}(8) = +1$

Boron: FC $= 3 - 0 - \frac{1}{2}(8) = -1$

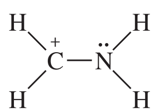
(d) $[H_2CNH_2]^+$



SOLUTION

In this structure, both carbon and nitrogen have four shared pairs of bonding electrons. With four bonds, carbon is formally neutral; however, nitrogen is in group 5A, and it bears a formal positive charge: FC $= 5 - 0 - 4 = +1$.

This compound might also be drawn with the following Lewis structure:



In this structure, the carbon atom has three bonds with six bonding electrons. We calculate that $\frac{6}{2} = 3$ electrons, so carbon is one short of the four needed to be formally neutral: FC $= 4 - 0 - \frac{1}{2}(6) = +1$.

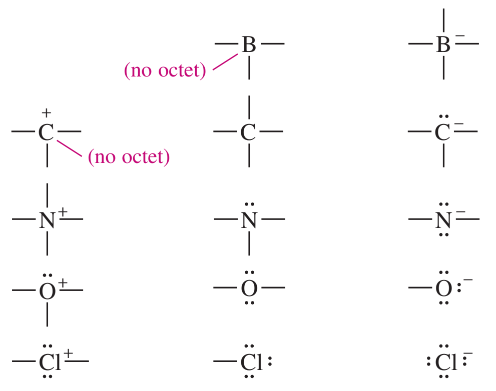
Nitrogen has six bonding electrons and two nonbonding electrons. We calculate that $\frac{6}{2} + 2 = 5$, so the nitrogen is uncharged in this second structure:

$$FC = 5 - 2 - \tfrac{1}{2}(6) = 0$$

The significance of these two Lewis structures is discussed in Section 1-9.

SUMMARY TABLE 1-1 Common Bonding Patterns in Organic Compounds and Ions

Atom	Valence Electrons	Positively Charged	Neutral	Negatively Charged
B	3		(no octet)	
C	4	(no octet)		
N	5			
O	6			
halogens	7			

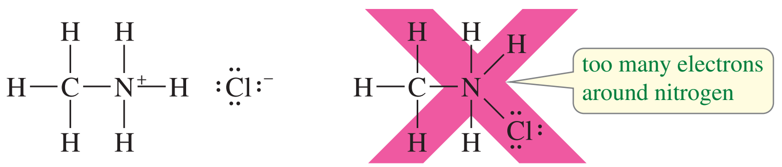


> **PROBLEM-SOLVING HINT**
>
> This is a very important table. Work enough problems to become familiar with these bonding patterns so that you can recognize other patterns as being either unusual or wrong.

Most organic compounds contain only a few common elements, usually with complete octets of electrons. Summary Table 1-1 shows the most commonly occurring bonding structures, using dashes to represent bonding pairs of electrons. Use the rules for calculating formal charges to verify the charges shown on these structures. A good understanding of the structures shown here will help you to draw organic compounds and their ions quickly and correctly.

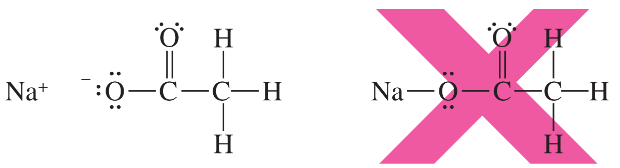
1-8 Ionic Structures

Some organic compounds contain ionic bonds. For example, the structure of methylammonium chloride (CH_3NH_3Cl) cannot be drawn using just covalent bonds. That would require nitrogen to have five bonds, implying ten electrons in its valence shell. The correct structure shows the chloride ion ionically bonded to the rest of the structure.



methylammonium chloride   cannot be drawn covalently

Some molecules can be drawn either covalently or ionically. For example, sodium acetate ($NaOCOCH_3$) may be drawn with either a covalent bond or an ionic bond between sodium and oxygen. Because sodium generally forms ionic bonds with oxygen (as in NaOH), the ionically bonded structure is usually preferred. In general, bonds between atoms with very large electronegativity differences (about 2 or more) are usually drawn as ionic.



drawn as ionic (more common)   drawn as covalent (less common)

PROBLEM 1-6

Draw Lewis structures for the following compounds and ions, showing appropriate formal charges.

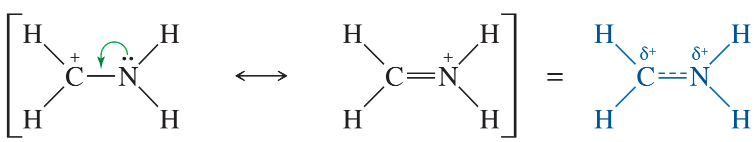
(a) $[CH_3OH_2]^+$ **(b)** NH_4Cl **(c)** $(CH_3)_4NCl$
(d) $NaOCH_3$ **(e)** $^+CH_3$ **(f)** $^-CH_3$
(g) $NaBH_4$ **(h)** $NaBH_3CN$ **(i)** $(CH_3)_2O—BF_3$
(j) $[HONH_3]^+$ **(k)** $KOC(CH_3)_3$ **(l)** $[H_2C{=}OH]^+$

1-9 Resonance

Some compounds' structures are not adequately represented by a single Lewis structure. When two or more valence-bond structures are possible, differing only in the placement of electrons, the molecule will usually show characteristics of both structures. The different structures are called **resonance structures** or **resonance forms** because they are not different compounds, just different ways of drawing the same compound. The actual molecule is said to be a **resonance hybrid** of its resonance forms.

1-9A Resonance Hybrids

In Solved Problem 1-1(d), we saw that the ion $[H_2CNH_2]^+$ might be represented by either of the following resonance forms:



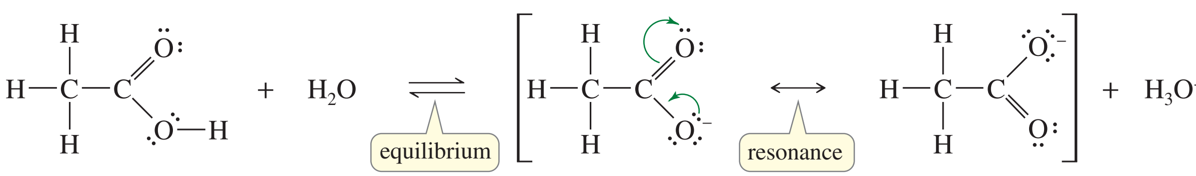
resonance forms of a resonance hybrid

combined representation

The actual structure of this ion is a resonance hybrid of the two structures. In the actual molecule, the positive charge is **delocalized** (spread out) over both the carbon atom and the nitrogen atom. In the left resonance form, the positive charge is on carbon, but carbon does not have an octet. We can imagine moving nitrogen's nonbonding electrons into the bond (as indicated by the green arrow) to give the second structure, with a positive charge on nitrogen and an octet on carbon. The combined representation attempts to combine the two resonance forms into a single picture, with the charge shared by carbon and nitrogen.

Spreading the positive charge over two atoms makes the ion more stable than it would be if the entire charge were localized only on the carbon or only on the nitrogen. We call this a **resonance-stabilized** cation. Resonance is most important when it allows a charge to be delocalized over two or more atoms, as in this example.

Resonance stabilization plays a crucial role in organic chemistry, especially in the chemistry of compounds having double bonds. For example, the acidity of acetic acid (shown below) is enhanced by resonance effects. When acetic acid loses a proton, the resulting acetate ion has a negative charge delocalized over both of the oxygen atoms. Each oxygen atom bears half of the negative charge, and this delocalization stabilizes the ion. Each of the carbon–oxygen bonds is halfway between a single bond and a double bond, and they are said to have a *bond order* of $1\frac{1}{2}$.

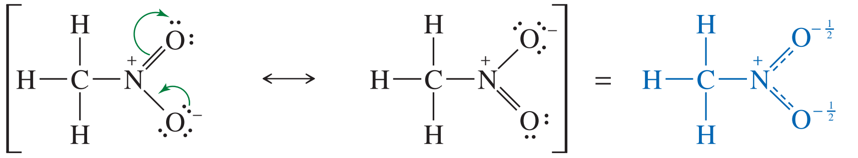


acetic acid

acetate ion

We use a single double-headed arrow between resonance forms (and often enclose them in brackets) to indicate that the actual structure is a hybrid of the Lewis structures we have drawn. By contrast, an equilibrium is represented by two arrows in opposite directions. Occasionally we use curved arrows (shown in green above) to help us see how we mentally move the electrons between one resonance form and another. The electrons do not actually move like these curved arrows show, and they do not "resonate" back and forth. They are delocalized over all the resonance forms at the same time.

Some uncharged molecules actually have resonance-stabilized structures with equal positive and negative formal charges. For example, we can draw two Lewis structures for nitromethane (CH_3NO_2), but both of them have a formal positive charge on nitrogen and a negative charge on one of the oxygens. Thus, nitromethane has a positive charge on the nitrogen atom and a negative charge spread equally over the two oxygen atoms. The N—O bonds are midway between single and double bonds, as indicated in the combined representation:



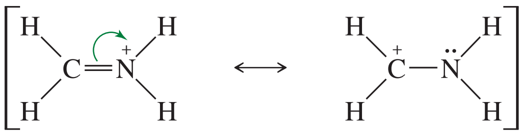
resonance forms

combined representation

Remember that individual resonance forms do not exist. The molecule does not "resonate" between these structures. It is a hybrid with some characteristics of both. An analogy is a mule, which is a hybrid of a horse and a donkey. The mule does not "resonate" between looking like a horse and looking like a donkey; it looks like a mule all the time, with the broad back of the horse and the long ears of the donkey.

1-9B Major and Minor Resonance Contributors

Two or more correct Lewis structures for the same compound may or may not represent electron distributions of equal energy. Although separate resonance forms do not exist, we can estimate their relative energies as if they did exist. More stable resonance forms are closer representations of the real molecule than less stable ones. The two resonance forms shown earlier for the acetate ion have similar bonding, and they are of identical energy. The same is true for the two resonance forms of nitromethane. The following resonance forms are bonded differently, however.



all octets (major contributor)

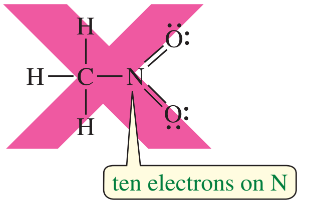
no octet on C (minor contributor)

These structures are not equal in estimated energy. The first structure has the positive charge on nitrogen. The second has the positive charge on carbon, and the carbon atom does not have an octet. The first structure is more stable because it has an additional bond and all the atoms have octets. Many stable ions have a positive charge on a nitrogen atom with four bonds (see Summary Table 1-1, page 42). We call the more stable resonance form the **major contributor,** and the less stable form is the **minor contributor.** Lower-energy (more stable) resonance forms are closer representations of the actual molecule or ion than are the higher-energy (less stable) ones.

Many organic molecules have major and minor resonance contributors. Formaldehyde ($H_2C{=}O$) can be written with a negative charge on oxygen, balanced by a positive charge on carbon. This polar resonance form is higher in estimated energy than the double-bonded structure because it has charge separation, fewer bonds, and

PROBLEM-SOLVING HINT

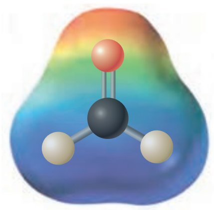
Second-row elements (B, C, N, O, F) cannot have more than eight electrons in their valence shells. The following is NOT a valid Lewis structure:



PROBLEM-SOLVING HINT

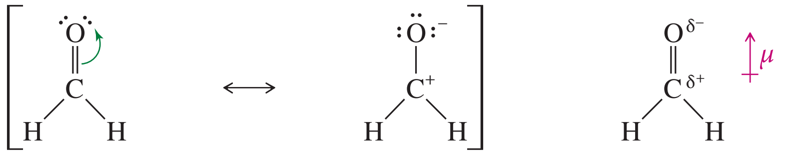
At this point, we are using four types of arrows in drawing structures and reactions:

- ⟶ Reaction arrows show the reactants converting to products.
- ⇌ Equilibrium arrows show equal rates of the forward and reverse reactions.
- ⟷ Resonance arrows connect forms that are actually the same structure but are drawn with the *delocalized* electrons arranged differently.
- Green curved arrows are a mental tool used to imagine moving electrons from one resonance form to another.



EPM of formaldehyde

a positively charged carbon atom without an octet. The charge-separated structure is only a minor contributor, but it helps to explain why the formaldehyde C═O bond is very polar, with a partial positive charge on carbon and a partial negative charge on oxygen. The electrostatic potential map (EPM) of formaldehyde also shows an electron-rich region (red) around oxygen and an electron-poor region (blue) around carbon.



all octets
more bonds
no charge separation
(major contributor)

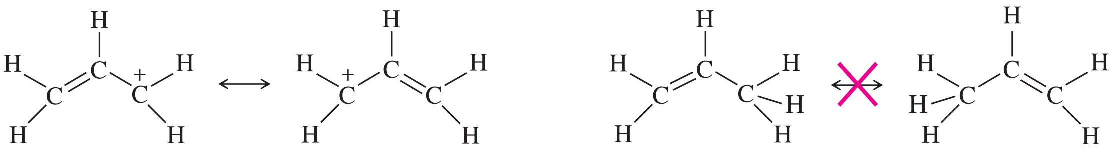
no octet on C
fewer bonds
charge separation
(minor contributor)

dipole moment

In drawing resonance forms, we try to draw structures that are as low in energy as possible. The best candidates are those that have the maximum number of octets and the maximum number of bonds. Also, we look for structures with the minimum amount of charge separation.

Only electrons can be delocalized. Unlike electrons, nuclei cannot be delocalized. They must remain in the same places, with the same bond distances and angles, in all the resonance contributors. The following general rules will help us to draw realistic resonance structures:

1. All the resonance forms must be valid Lewis structures for the compound. Second-row elements (B, C, N, O, F) can never have more than eight electrons in their valence shells.
2. Only the placement of the electrons may be shifted from one structure to another. Electrons in double bonds and nonbonding electrons (lone pairs) are most commonly shifted.
3. Nuclei cannot be moved, and all bond angles must remain the same.



resonance forms

NOT resonance
a H nucleus has moved

PROBLEM-SOLVING HINT

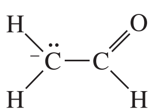
Resonance forms can be compared using the following criteria, beginning with the most important:

1. As many octets as possible
2. As many bonds as possible
3. As little charge separation as possible
4. Any negative charges on electronegative atoms

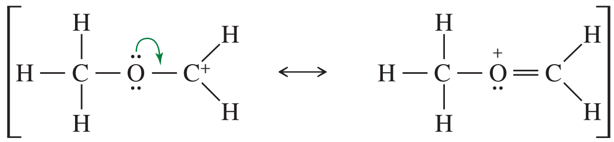
4. Sigma bonds are very stable, and they are rarely involved in resonance.
5. The major resonance contributor is the one with the lowest energy. The best contributors generally have the most octets satisfied, as many bonds as possible, and as little charge separation as possible.
6. Electronegative atoms such as N, O, and halogens often help to delocalize positive charges, but they can bear a positive charge only if they have octets.
7. Resonance stabilization is most important when it serves to delocalize a charge or a radical over two or more atoms.

SOLVED PROBLEM 1-2

For each of the following compounds, draw the important resonance forms. Indicate which structures are major and minor contributors or whether they would have the same energy.

(a) $[CH_3OCH_2]^+$ **(b)**  **(c)** H_2SO_4

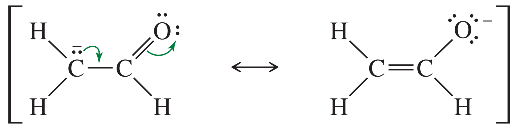
SOLUTION (a)



minor contributor major contributor

The first (minor) structure has a carbon atom with only six electrons around it. The second (major) structure has octets on all atoms and an additional bond.

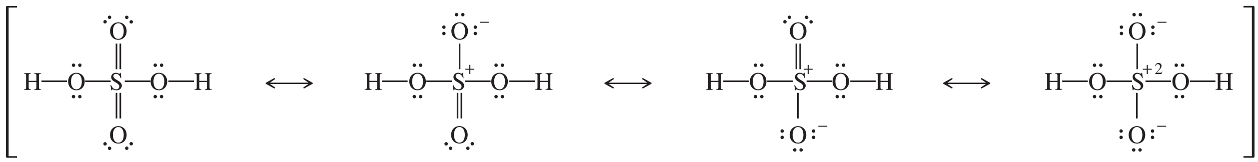
SOLUTION (b)



minor contributor major contributor

Both of these structures have octets on oxygen and both carbon atoms, and they have the same number of bonds. The first structure has the negative charge on carbon; the second has it on oxygen. Oxygen is the more electronegative element, so the second structure is the major contributor.

SOLUTION (c)



The first structure, with more bonds and less charge separation, is possible because sulfur is a third-row element with accessible *d* orbitals, giving it an expandable valence. For example, SF_6 is a stable compound with 12 electrons around sulfur. Theoretical calculations suggest that the last structure, with octets on all atoms, may be the major resonance contributor, however. Organic chemists mostly draw the first uncharged structure, whereas inorganic chemists mostly prefer the fourth structure with octets on all atoms.

PROBLEM 1-7

Draw the important resonance forms for the following molecules and ions.

(a) CO_3^{2-} **(b)** $H_2C{=}CH{-}\overset{+}{C}H_2$ **(c)** $H_2C{=}CH{-}\overset{-}{C}H_2$ **(d)** NO_3^-

(e) NO_2^- **(f)** $H{-}\overset{\overset{\large O}{\|}}{C}{-}CH{=}CH{-}\overset{-}{C}H_2$ **(g)** $[CH_3C(OCH_3)_2\,]^+$ **(h)** $B(OH)_3$

PROBLEM-SOLVING HINT

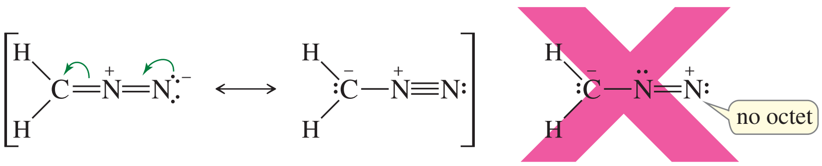
In drawing resonance forms for ions, see how you can delocalize the charge over several atoms.

Try to spread a negative charge over electronegative elements such as oxygen and nitrogen. Try to spread a positive charge over as many carbons as possible, but especially over any atoms that can bear the positive charge and still have an octet, such as oxygen (with three bonds) or nitrogen (with four bonds).

PROBLEM 1-8 (PARTIALLY SOLVED)

For each of the following compounds, draw the important resonance forms. Indicate which structures are major and minor contributors or whether they have the same energy.

(a) H_2CNN



major minor not significant

(continued)

SOLUTION

The first two resonance forms have all their octets satisfied, and they have the same number of bonds. They show that the central nitrogen atom bears a positive charge, and the outer carbon and nitrogen share a negative charge. A nitrogen atom is more electronegative than a carbon atom, so we expect the structure showing the minus charge on nitrogen to be the major contributor.

The third structure is not one we would normally draw, because it is much less significant than the other two. It has a nitrogen atom without an octet, and it has fewer bonds than the other two structures. Nitrogen often bears a positive charge when it has four bonds and an octet, but it rarely bears a positive charge without an octet.

(b) $[H_2CCN]^-$ **(c)** $[H_2COCH_3]^+$ **(d)** $[H_2CNO_2]^-$ **(e)** $[CH_3C(OH)_2]^+$

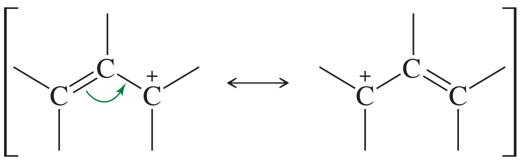
(f) $[CH_2CHNH]^-$ **(g)** H—C(=O)—$\overset{-}{\text{C}}$H—C(=O)—H **(h)** CHO—$\overset{-}{\text{C}}$H—CN

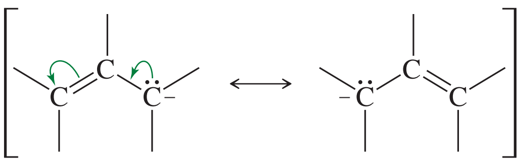
PROBLEM-SOLVING STRATEGY Drawing and Evaluating Resonance Forms

We commonly draw resonance forms (resonance structures) for one of two reasons:

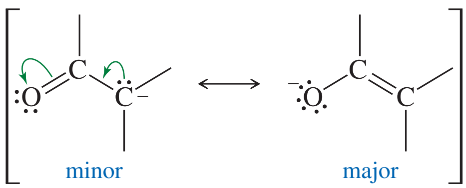
1. To show how delocalization stabilizes a reactive species, such as a cation or anion, by spreading the charge over two or more atoms in a structure.
2. To explain the characteristics of a compound and why it reacts the way it does.

In drawing resonance forms to show how a cation or an anion is delocalized, you should look for double bonds and nonbonding electron pairs (lone pairs) next to the charged atom. A double bond next to a charged atom can delocalize the charge. In drawing resonance forms, you can use curved arrows to keep track of the electrons as you mentally move them from one place to another. Remember that this movement is imaginary. The actual molecule is a mixture of all the resonance forms all the time. The electrons and charges do not move back and forth, but are delocalized throughout the structure.

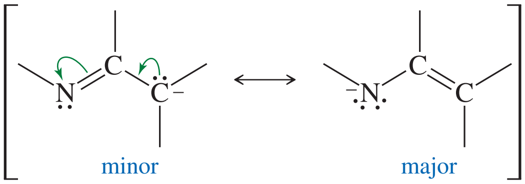




If the double bond allows an electronegative element to share a negative charge, that resonance form will be particularly stable. The most common examples are double bonds to oxygen and nitrogen.



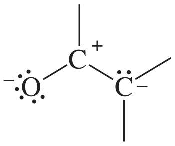
The major resonance form has the negative charge on oxygen, an electronegative element.



The major resonance form has the negative charge on nitrogen, an electronegative element.

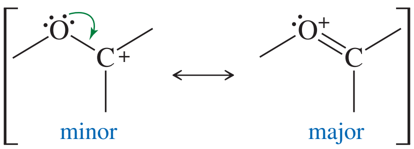
These resonance forms show how the negative charge is spread over a carbon atom and the oxygen or nitrogen atom. All the atoms have octets in both structures, so the more important resonance form is the one with the negative charge on the more electronegative element, either oxygen or nitrogen. Still, these resonance forms show why this anion can react at either the carbon or the oxygen or nitrogen atom.

The resonance form shown below is also a valid Lewis structure, as well as a valid resonance form, but it is not an important resonance form (except in quantum-mechanical calculations) because it does not help to delocalize the negative charge, and it includes unnecessary charge separation and a carbon atom without an octet. You would not draw this resonance form in most cases.

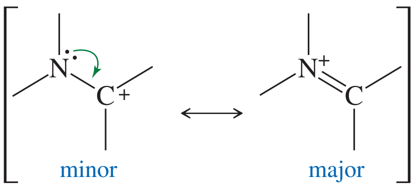


Not an important resonance form. Too much charge separation, too few bonds, and a carbon atom without an octet.

A carbon atom with a positive charge lacks an octet. If it is bonded to an atom with nonbonding electrons (often oxygen or nitrogen), the carbon atom can share those nonbonding electrons. This sharing allows it to gain an octet and delocalize the positive charge onto its neighbor. The following examples show how this delocalization affords more stability to the cation.



The major resonance form has all octets satisfied, and it has more bonds than the other form.



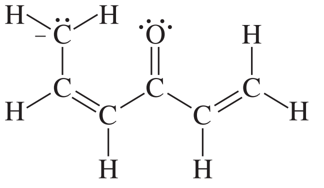
The major resonance form has all octets satisfied, and it has more bonds than the other form.

At first, it seems surprising that the major resonance form has the positive charge on oxygen or nitrogen, which are both more electronegative than carbon. But here, the comparison is between carbon with a positive charge and lacking an octet, versus oxygen or nitrogen with a positive charge and a complete octet. This situation is similar to the stable ions H_3O^+ and NH_4^+, which also bear a positive charge on oxygen or nitrogen, but with a complete octet.

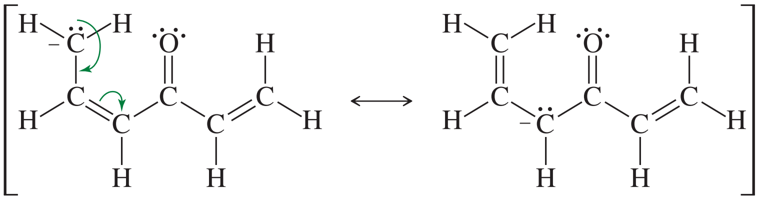
Finally, let's see how you would approach problems that involve delocalizing positive and negative charges.

SOLVED PROBLEM 1-3

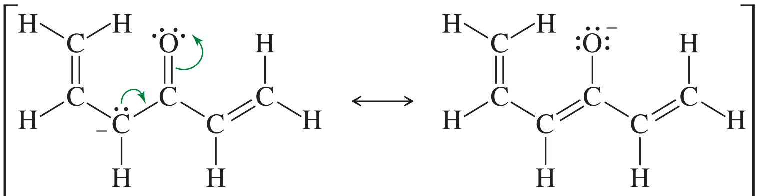
Draw the important resonance forms of the following anion:



Looking ahead, you should try to put the negative charge on the electronegative oxygen atom of the C═O group. But first, the carbon with the negative charge has a C═C double bond next to it. The lone pair can delocalize into the double bond, sharing the negative charge with another carbon.



This second structure now has a C═O double bond next to the carbon bearing the negative charge. Moving the pair of electrons in that direction delocalizes the negative charge onto the oxygen atom, a favorable result.

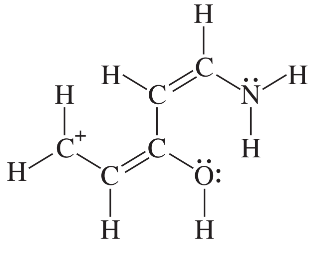


major, negative charge on O

Note that you cannot delocalize the negative charge into the final double bond on the right. There are no valid Lewis structures that place the negative charge on either of these two carbon atoms.

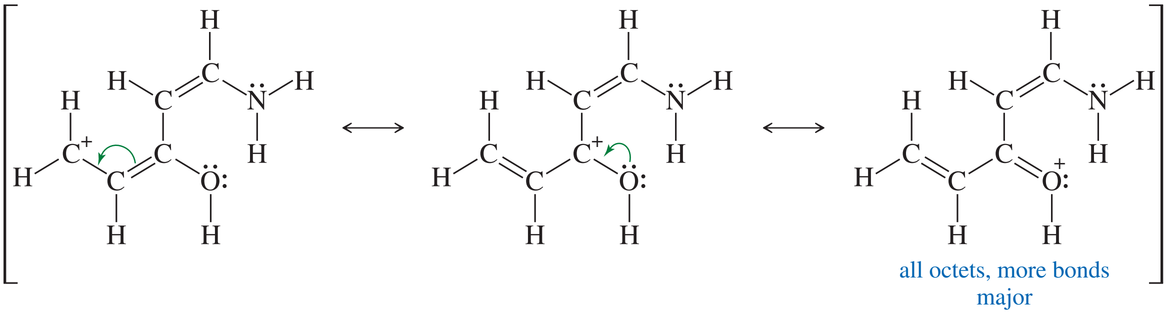
SOLVED PROBLEM 1-4

Draw the important resonance forms of the following cation:

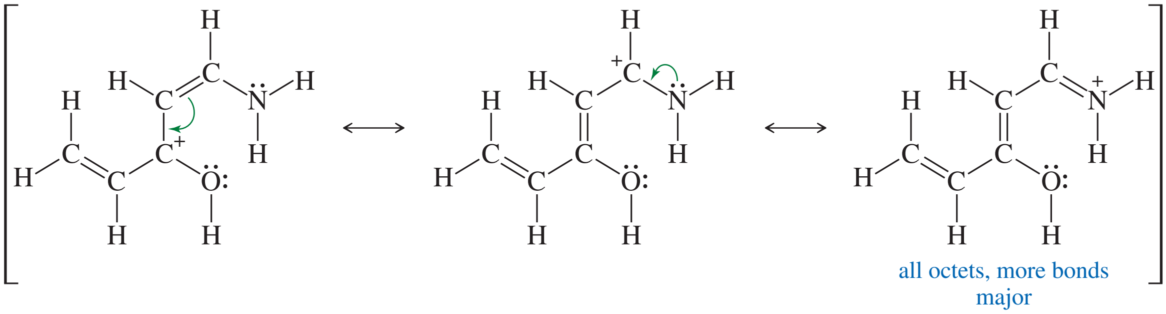


SOLUTION

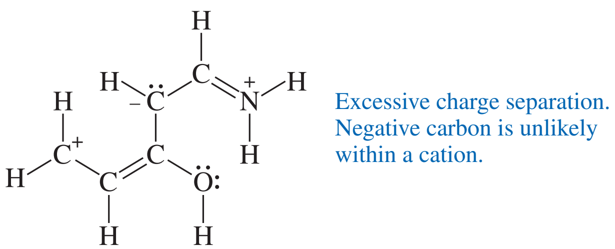
Looking ahead, you should try to put the positive charge on oxygen (with three bonds) or nitrogen (with four bonds) to give resonance forms with octets. But first, notice the double bond next to the carbon with the positive charge. Moving the electrons in that double bond shares the charge with another carbon. This second carbon has a neighbor (oxygen) with lone pairs that can provide carbon with an octet, giving a relatively stable structure.



This second carbon also has another double bond next to it. Moving the electrons in this double bond moves the charge to a third carbon. The third carbon with the positive charge has a neighbor (nitrogen) with lone pairs that can provide carbon with an octet, giving another relatively stable structure.



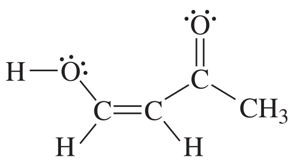
These are all of the important resonance forms that you would normally draw. You could draw other valid Lewis structures, like the one below, but they do not help to delocalize the charge, and they have unnecessary charge separation. These are not important resonance forms. This one also shows a negative charge on a carbon atom, which is unrealistic in a cation where we seek to delocalize a positive charge. You would not normally draw this resonance form.



Finally, resonance forms can help you to predict where you should expect high or low electron density in a molecule. To do that, look for a highly polar area and see how resonance delocalizes those partial charges over the molecule. Atoms that bear a positive charge in one or more resonance forms are likely to be electron-poor, while those that bear a negative charge are likely to be electron-rich.

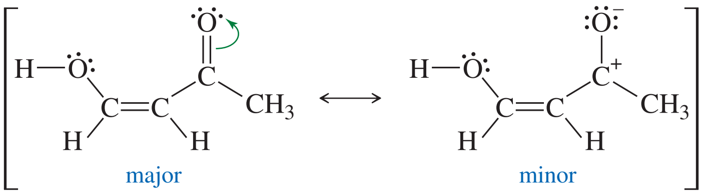
SOLVED PROBLEM 1-5

Use resonance structures to identify the areas of high and low electron density in the following compound:

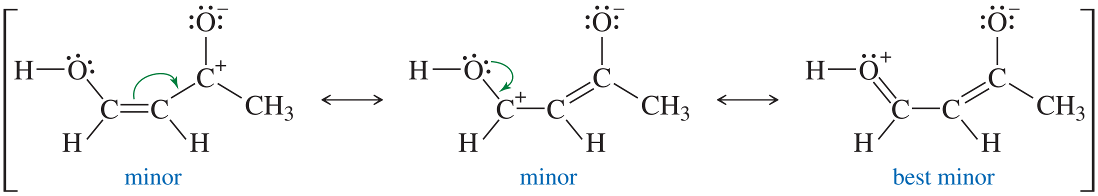


The C=O bond is highly polar, so you could begin by showing charges to represent its polarity. This resonance form is not nearly as stable as the original uncharged structure, but it helps to visualize the partial charges.

(*continued*)



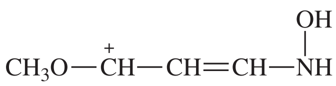
Moving the bonds and nonbonding electrons next to these charges gives other important resonance forms. Note how the oxygen of the OH group is an electron donor, while the oxygen of the C═O group is an electron acceptor.

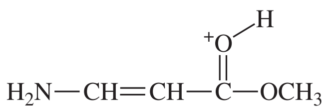


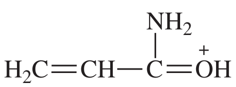
The final resonance form ("best minor") is the most stable of the structures with charge separation. The original uncharged structure labeled "major" is the best of all, with no charge separation, all octets satisfied, and the maximum number of bonds. It is more representative of the actual structure than any of the charge-separated resonance forms. Still, the charge-separated forms suggest where the molecule is likely to have electron-rich (negative) and electron-poor (positive) regions. We predict that the OH oxygen at left and the first and third carbons are electron-poor, while the oxygen at right (from the C═O double bond) is electron-rich.

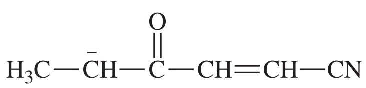
PROBLEM 1-9

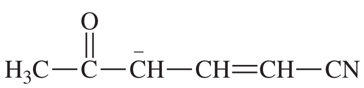
Draw the important resonance forms of the following cations and anions:

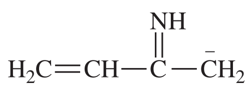
(a)


(b)


(c)


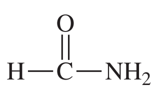
(d)


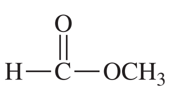
(e)


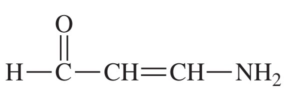
(f)


PROBLEM 1-10

Use resonance structures to identify the areas of high and low electron density in the following compounds:

(a) $H_2C{=}CH{-}NO_2$ **(b)** $H_2C{=}CH{-}C{\equiv}N$ **(c)**


(d)


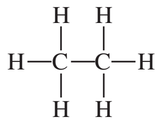
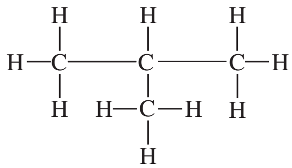
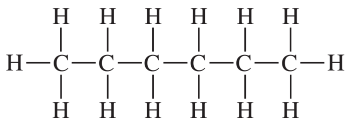
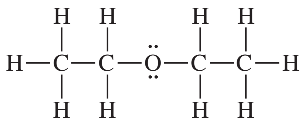
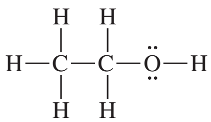
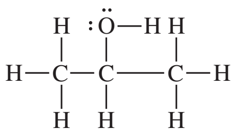
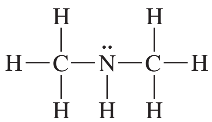
(e)


(f) $CH_3O{-}CH{=}CH{-}CN$

1-10 Structural Formulas

Organic chemists use several kinds of formulas to represent organic compounds. Some of these formulas involve a shorthand notation that requires some explanation. **Structural formulas** actually show which atoms are bonded to which. Structural formulas are of

TABLE 1-2 Examples of Condensed Structural Formulas

Compound	Lewis Structure	Condensed Structural Formula
ethane		CH_3CH_3
isobutane		$(CH_3)_3CH$
n-hexane		$CH_3(CH_2)_4CH_3$
diethyl ether		$CH_3CH_2OCH_2CH_3$ or $CH_3CH_2{-}O{-}CH_2CH_3$ or $(CH_3CH_2)_2O$
ethanol		CH_3CH_2OH
isopropyl alcohol		$(CH_3)_2CHOH$
dimethylamine		$(CH_3)_2NH$

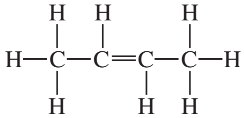
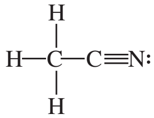
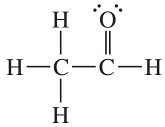
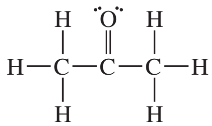
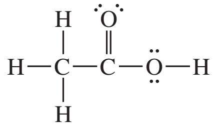
two major types: complete Lewis structures and condensed structural formulas. In addition, there are several ways of drawing condensed structural formulas. As we have seen, a Lewis structure symbolizes a bonding pair of electrons as a pair of dots or as a dash (—). Lone pairs of electrons are shown as pairs of dots.

1-10A Condensed Structural Formulas

Condensed structural formulas (Table 1-2) are written without showing all the individual bonds. In a condensed structural formula, each central atom is shown together with the atoms that are bonded to it. The atoms bonded to a central atom are often listed after the central atom (as in CH_3CH_3 rather than $H_3C{-}CH_3$) even if that is not their actual bonding order. In many cases, if there are two or more identical groups, parentheses and a subscript may be used to represent all the identical groups. Nonbonding electrons are rarely shown in condensed structural formulas.

When we write a condensed structural formula for a compound containing double or triple bonds, the multiple bonds are often drawn as they would be in a Lewis structure. Table 1-3 shows examples of condensed structural formulas containing multiple bonds. Notice that the —CHO group of an aldehyde and the —COOH group of a carboxylic acid are actually bonded differently from what the condensed notation suggests. Condensed structures are assumed to follow the octet rule even if the condensed notation does not show the bonding.

TABLE 1-3 Condensed Structural Formulas for Double and Triple Bonds

Compound	Lewis Structure	Condensed Structural Formula
but-2-ene		$CH_3CHCHCH_3$ or $CH_3CH{=}CHCH_3$
acetonitrile		CH_3CN or $CH_3C{\equiv}N$
acetaldehyde		CH_3CHO or $CH_3\overset{\displaystyle O}{\overset{\|}{C}}H$
acetone		CH_3COCH_3 or $CH_3\overset{\displaystyle O}{\overset{\|}{C}}CH_3$
acetic acid		CH_3COOH or $CH_3\overset{\displaystyle O}{\overset{\|}{C}}OH$ or CH_3CO_2H

As you can see from Tables 1-2 and 1-3, the distinction between a complete Lewis structural formula and a condensed structural formula can be blurry. Chemists often draw formulas with some parts condensed and other parts completely drawn out. You should work with these different types of formulas so that you understand what all of them mean.

PROBLEM 1-11

Draw complete Lewis structures for the following condensed structural formulas.

(a) $CH_3(CH_2)_3CH(CH_3)_2$
(b) $(CH_3)_2CHCH_2Cl$
(c) CH_3CH_2COCN
(d) CH_2CHCHO
(e) $(CH_3)_3CCOCHCH_2$
(f) $CH_3COCOOH$
(g) $(CH_3CH_2)_2CO$
(h) $(CH_3)_3COH$

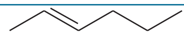
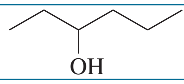
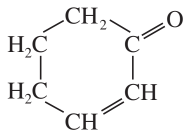
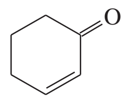
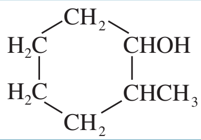
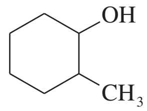
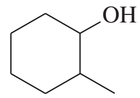
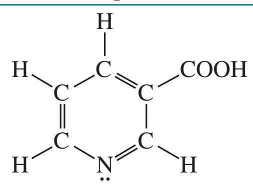
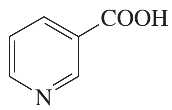
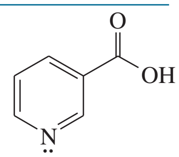
PROBLEM-SOLVING HINT

In a line–angle formula, a carbon atom is implied at the end of every line and at every vertex, unless another atom is specified.

1-10B Line–Angle Formulas

Another kind of shorthand used for organic structures is the **line–angle formula**, sometimes called a **skeletal structure** or a **stick figure.** Line–angle formulas are often used for cyclic compounds and occasionally for noncyclic ones. In a stick figure, bonds are represented by lines, and carbon atoms are assumed to be present wherever two lines meet or a line begins or ends. Nitrogen, oxygen, and halogen atoms are shown, but hydrogen atoms are not usually drawn unless they are bonded to an atom that is drawn. Each carbon atom is assumed to have enough hydrogen atoms to give it a total of four bonds. Nonbonding electrons are rarely shown. Table 1-4 shows some examples of line–angle drawings.

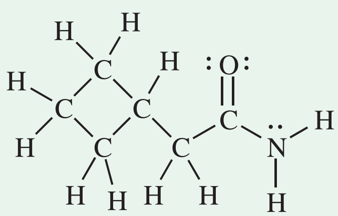
TABLE 1-4 Examples of Line–Angle Drawings

Compound	Condensed Structure	Line–Angle Formula
hexane	$CH_3(CH_2)_4CH_3$	
hex-2-ene	$CH_3CH{=}CHCH_2CH_2CH_3$	
hexan-3-ol	$CH_3CH_2CH(OH)CH_2CH_2CH_3$	
cyclohex-2-en-1-one		
2-methylcyclohexan-1-ol (2-methylcyclohexanol)		 or 
nicotinic acid (a vitamin, also called niacin)		 or 

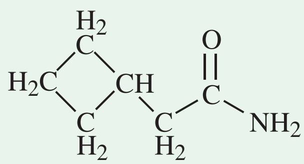
FOCUS Drawing Structures

Chemists represent organic compounds not just by their molecular formula, but by a graphical depiction of how the atoms are connected. Early in your study of organic chemistry, you will be expected to master this graphic language. We use different types of structural representations, with varying degrees of simplification.

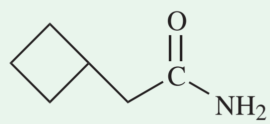
A full structural formula, also called a **Lewis structure**, shows all of the atoms, all of the bonds, and all of the unshared (lone) pairs of electrons.



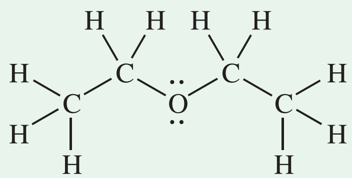
A **condensed structural formula** shows all of the atoms and some of the bonds; bonds to H are usually omitted. In a carbon chain, the bonds between carbons are usually not shown. Unshared (lone) pairs of electrons may or may not be shown.



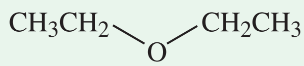
A line–angle formula, or simply a **line formula**, has the most simplification. H atoms and C atoms are usually not shown unless bonded to an atom other than C.



This example of diethyl ether (commonly called "ether") shows the three types of representations.

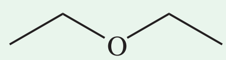


Lewis structure or full structural formula



or $CH_3CH_2OCH_2CH_3$

condensed structural formula



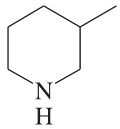
line–angle formula or line formula

Note that these structural representations are not intended to show accurate bond angles or positions of the atoms in three dimensions. In later chapters, we will introduce different types of notation to display accurate three-dimensional positions of atoms.

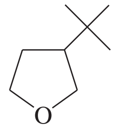
PROBLEM 1-12

Give Lewis structures corresponding to the following line–angle structures. Give the molecular formula for each structure.

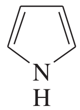
(a)



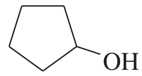
(b)



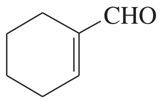
(c)



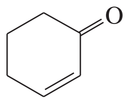
(d)



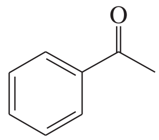
(e)



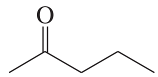
(f)



(g)



(h)



PROBLEM 1-13

Repeat Problem 1-11, this time drawing line–angle structures for compounds (a) through (h).

1-11 Molecular Formulas and Empirical Formulas

Before we can write possible structural formulas for a compound, we need to know its molecular formula. The **molecular formula** simply gives the number of atoms of each element in one molecule of the compound. For example, the molecular formula for butan-1-ol is $C_4H_{10}O$.

$$CH_3CH_2CH_2CH_2OH$$

butan-1-ol, molecular formula $C_4H_{10}O$

Calculation of the Empirical Formula Molecular formulas can be determined by a two-step process. The first step is the determination of an **empirical formula,** simply the relative ratios of the elements present. Suppose, for example, that an unknown compound was found by quantitative elemental analysis to contain 40.0% carbon and 6.67% hydrogen. The remainder of the weight (53.3%) is assumed to be oxygen. To convert these numbers to an empirical formula, we can follow a simple procedure.

1. Assume that the sample contains 100 g, so the percent value gives the number of grams of each element. Divide the number of grams of each element by the atomic weight to get the number of moles of that atom in the 100-g sample.
2. Divide each of these numbers of moles by the smallest one. This step should give recognizable ratios.

   For the unknown compound, we do the following computations:

$$\frac{40.0\text{ g C}}{12.0\text{ g/mol}} = 3.33\text{ mol C}; \qquad \frac{3.33\text{ mol}}{3.33\text{ mol}} = 1$$

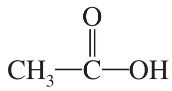
$$\frac{6.67\text{ g H}}{1.01\text{ g/mol}} = 6.60\text{ mol H}; \qquad \frac{6.60\text{ mol}}{3.33\text{ mol}} = 1.98 \cong 2$$

$$\frac{53.3\text{ g O}}{16.0\text{ g/mol}} = 3.33\text{ mol O}; \qquad \frac{3.33\text{ mol}}{3.33\text{ mol}} = 1$$

The first computation divides the number of grams of carbon by 12.0, the number of grams of hydrogen by 1.0, and the number of grams of oxygen by 16.0. We compare the number of moles of C, H, and O by dividing them by the smallest number, 3.33. The final result is a ratio of one carbon to two hydrogens to one oxygen. This result gives the empirical formula $C_1H_2O_1$ or CH_2O, which simply shows the ratios of the elements. The molecular formula can be any multiple of this empirical formula, because any multiple also has the same ratio of elements. Possible molecular formulas are CH_2O, $C_2H_4O_2$, $C_3H_6O_3$, $C_4H_8O_4$, etc.

Calculation of the Molecular Formula How do we know the correct molecular formula? We can choose the right multiple of the empirical formula if we know the molecular weight. Molecular weights can be determined by methods that relate the freezing-point depression or boiling-point elevation of a solvent to the molal concentration of the unknown. If the compound is volatile, we can convert it to a gas and use its volume to determine the number of moles according to the gas law. Newer methods for determining molecular weights include *mass spectrometry*, which we will cover in Chapter 12.

For our example (empirical formula CH_2O), let's assume that the molecular weight is determined to be about 60. The weight of one CH_2O unit is 30, so our unknown compound must contain twice this many atoms. The molecular formula must be $C_2H_4O_2$. The compound might be acetic acid or any of several other compounds with this molecular formula.



acetic acid, $C_2H_4O_2$

In Chapters 12, 13, and 15 we will use spectroscopic techniques to determine the complete structure for a compound once we know its molecular formula.

PROBLEM 1-14

Compute the empirical and molecular formulas for each of the following elemental analyses. In each case, propose at least one structure that fits the molecular formula.

	C	H	N	Cl	MW
(a)	40.0%	6.67%	0	0	90
(b)	32.0%	6.67%	18.7%	0	75
(c)	25.6%	4.32%	15.0%	37.9%	93
(d)	38.4%	4.80%	0	56.8%	125

PROBLEM-SOLVING HINT

If an elemental analysis does not add up to 100%, the missing percentage is assumed to be oxygen.

1-12 Wave Properties of Electrons in Orbitals

The chemistry we have covered so far does not explain the actual shapes and properties of organic molecules. To understand these aspects of molecular structure, we need to consider how the atomic orbitals on an atom mix to form *hybrid atomic orbitals* and how orbitals on different atoms combine to form *molecular orbitals*. Now we consider how combinations of orbitals account for the shapes and properties we observe in organic molecules.

1-12A Waveforms and Nodes

We like to picture the atom as a miniature solar system, with the electrons orbiting around the nucleus. This solar system picture satisfies our intuition, but it does not accurately reflect today's understanding of the atom. About 1923, Louis de Broglie suggested that the properties of electrons in atoms are better explained by treating the electrons as waves rather than as particles.

There are two general kinds of waves, *traveling waves* and *standing waves*. Examples of traveling waves are the sound waves that carry a thunderclap and the water waves that form the wake of a boat. Standing waves vibrate in a fixed location. Standing waves are found inside an organ pipe, where the rush of air creates a vibrating air column, and in the wave pattern of a guitar string when it is plucked. An electron in an atomic orbital is like a stationary, bound vibration: a standing wave.

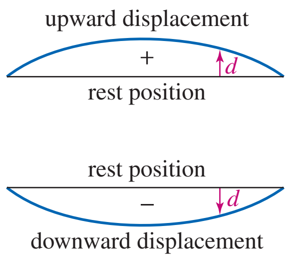


FIGURE 1-9 A standing wave. The fundamental frequency of a guitar string is a standing wave with the string alternately displaced upward and downward.

If we consider an electron in an orbital as a three-dimensional standing wave, what are the resulting properties? Let's first consider a one-dimensional analog of this standing wave; namely, the vibration of a guitar string (Figure 1-9). If you pluck a guitar string at its middle, a standing wave results. In this mode of vibration, all of the string

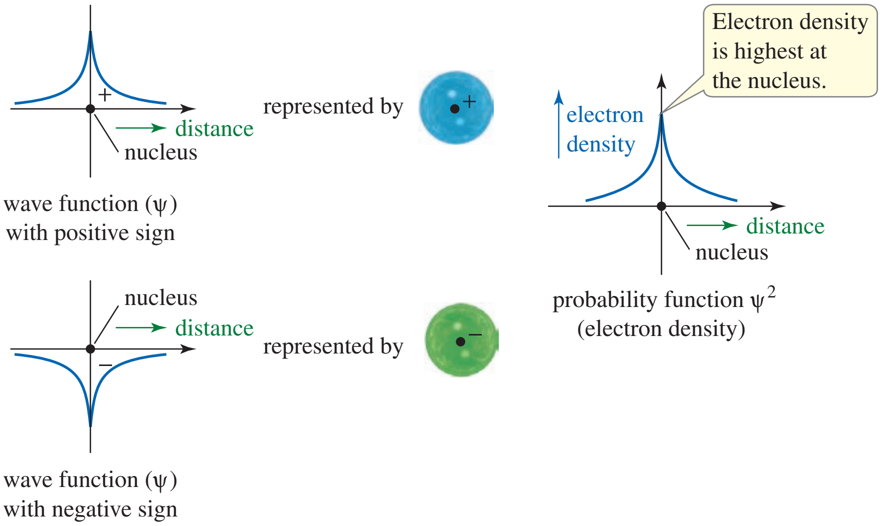


FIGURE 1-10 The 1*s* orbital. The 1*s* orbital is similar to the fundamental vibration of a guitar string. The wave function is instantaneously all positive or all negative. The square of the wave function gives the electron density. A circle with a nucleus is used to represent the spherically symmetrical *s* orbital.

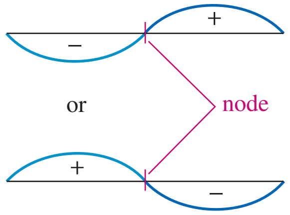


FIGURE 1-11 First harmonic of a guitar string. The two halves of the string are separated by a node, a point with zero displacement. The two halves vibrate out of phase with each other.

is displaced upward for a fraction of a second, then downward for an equal time. An instantaneous picture of the waveform shows the string displaced in a smooth curve either upward or downward, depending on the exact instant of the picture.

The waveform of a 1*s* orbital is like this guitar string, except that it is three-dimensional. The orbital can be described by its **wave function**, ψ, which is the mathematical description of the shape of the wave as it vibrates. All of the wave is positive in sign for a brief instant; then it is negative in sign. The electron density at any point is given by ψ^2, the square of the wave function at that point. *The plus sign and the minus sign of these wave functions are not charges. The plus or minus sign is the instantaneous phase of the constantly changing wave function*. The 1*s* orbital is spherically symmetrical, and it is often represented by a circle (representing a sphere) with a nucleus in the center and with a plus or minus sign to indicate the instantaneous sign of the wave function (Figure 1-10).

If you gently place a finger at the center of a guitar string while plucking the string, your finger keeps the midpoint of the string from moving. The displacement (movement + or –) at the midpoint is always zero; this point is a **node.** The string now vibrates in two parts, with the two halves vibrating in opposite directions. We say that the two halves of the string are *out of phase:* When one is displaced upward, the other is displaced downward. Figure 1-11 shows this first *harmonic* of the guitar string.

The first harmonic of the guitar string resembles the 2*p* orbital (Figure 1-12). We have drawn the 2*p* orbital as two "lobes," separated by a node (a nodal plane). The two lobes of the *p* orbital are out of phase with each other. Whenever the wave function has

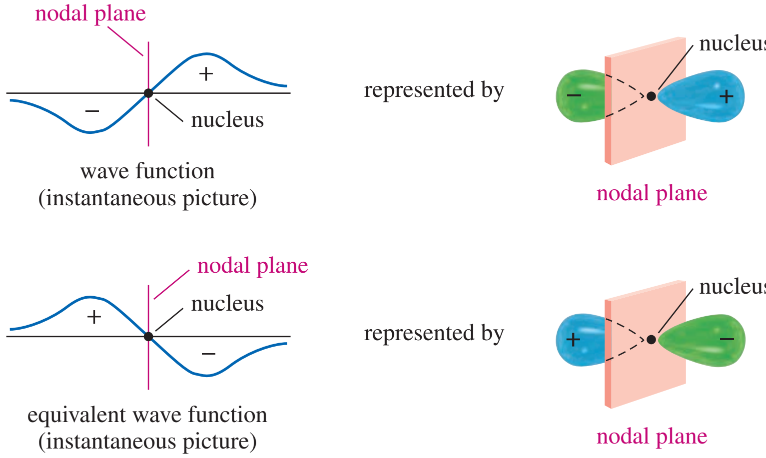


FIGURE 1-12 The 2*p* orbital. The 2*p* orbital has two lobes separated by a nodal plane. The two lobes are out of phase with each other. When one has a plus sign, the other has a minus sign.

a plus sign in one lobe, it has a minus sign in the other lobe. When phase relationships are important, organic chemists often represent the phases with colors. Figures 1-10 and 1-12 use blue for regions with a positive phase and green for a negative phase.

1-12B Linear Combination of Atomic Orbitals

Atomic orbitals can combine and overlap to give more complex standing waves. We can add and subtract their wave functions to give the wave functions of new orbitals. This process is called **linear combination of atomic orbitals** (LCAO). The number of new orbitals generated always equals the number of starting orbitals.

1. When orbitals on *different* atoms interact, they produce **molecular orbitals** (MOs) that lead to bonding (or antibonding) interactions.
2. When orbitals on the *same* atom interact, they give **hybrid atomic orbitals** that define the geometry of the bonds.

We begin by looking at how atomic orbitals on different atoms interact to give molecular orbitals. Then we consider how atomic orbitals on the same atom can interact to give hybrid atomic orbitals.

PROBLEM-SOLVING HINT

When orbitals combine to form hybrid atomic orbitals or molecular orbitals, the number of orbitals formed always equals the number of orbitals that combine to form them.

1-13 Molecular Orbitals

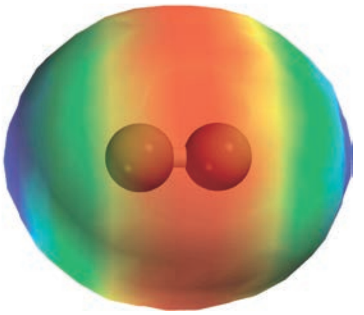
The stability of a covalent bond results from a large amount of electron density in the bonding region, the space between the two nuclei (Figure 1-13). In the bonding region, the electrons are close to both nuclei, lowering the overall energy. The bonding electrons also mask the positive charges of the nuclei, so the nuclei do not repel each other as much as they would otherwise.

There is always an optimum distance for the two bonded nuclei. If they are too far apart, their attraction for the bonding electrons is diminished. If they are too close together, their electrostatic repulsion pushes them apart. The internuclear distance where attraction and repulsion are balanced, which also gives the minimum energy (the strongest bond), is the *bond length*.

1-13A The Hydrogen Molecule; Sigma Bonding

The hydrogen molecule is the simplest example of covalent bonding. As two hydrogen atoms approach each other, their 1*s* wave functions can add *constructively* so that they reinforce each other or *destructively* so that they cancel out where they overlap. Figure 1-14 shows how the wave functions interact constructively when they are in phase and have the same sign in the region between the nuclei. The wave functions reinforce each other and increase the electron density in this bonding region. The result is a **bonding molecular orbital** (bonding MO).

The bonding MO depicted in Figure 1-14 has most of its electron density centered *along the line connecting the nuclei*. This type of bond is called a *cylindrically symmetrical bond* or a **sigma bond (σ bond).** Sigma bonds are the most common bonds in organic compounds. All single bonds in organic compounds are sigma bonds, and every double or triple bond contains one sigma bond. The electrostatic potential map



EPM of H_2

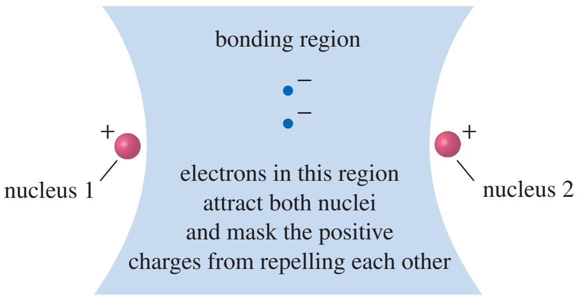


FIGURE 1-13 The bonding region. Electrons in the space between the two nuclei attract both nuclei and mask their positive charges. A bonding molecular orbital places a large amount of electron density in the bonding region.

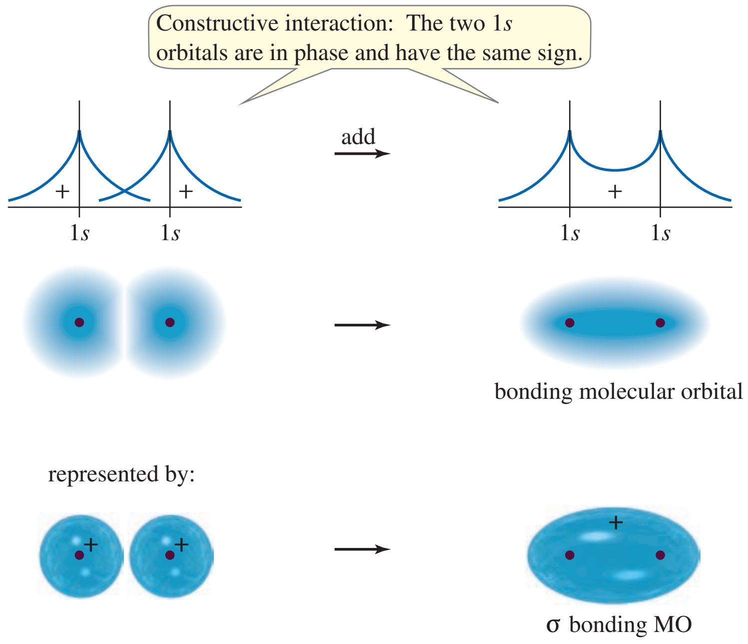


FIGURE 1-14 Formation of a σ bonding MO. When the 1s orbitals of two hydrogen atoms overlap in phase, they interact constructively to form a bonding MO. The electron density in the bonding region (between the nuclei) is increased. The result is a cylindrically symmetrical bond, or sigma (σ) bond.

(EPM) of H_2 shows its cylindrically symmetrical sigma bond, with the highest electron density (red) in the bonding region between the two protons.

When two hydrogen 1s orbitals overlap *out of phase* with each other, an **antibonding molecular orbital** results (Figure 1-15). The two 1s wave functions have opposite signs, so they tend to cancel out where they overlap. The result is a node (actually a nodal plane) separating the two atoms. The presence of a node separating the two nuclei

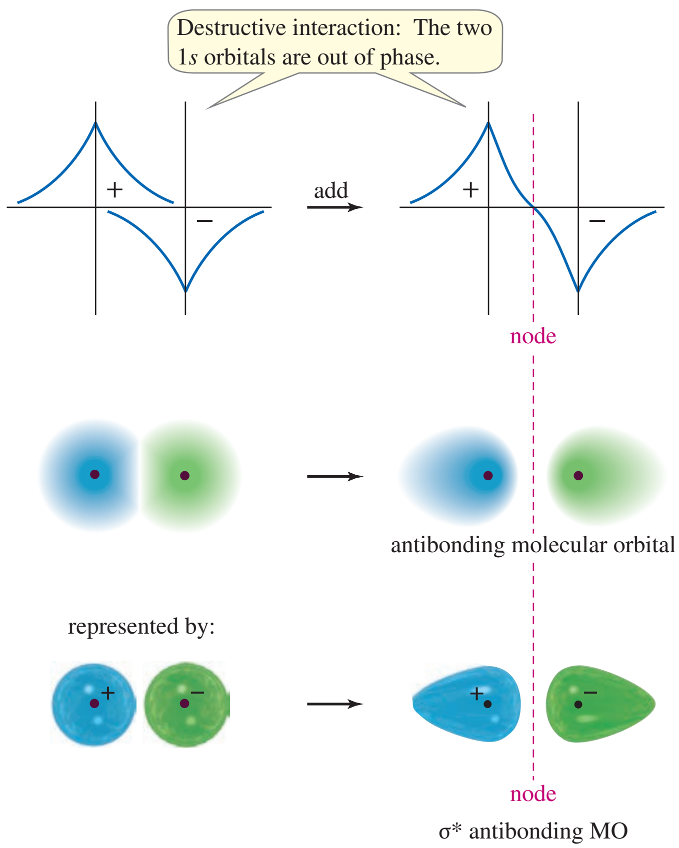


FIGURE 1-15 Formation of a σ^* antibonding MO. When two 1s orbitals overlap out of phase, they interact destructively to form an antibonding MO. The positive and negative values of the wave functions tend to cancel out in the region between the nuclei, and a node separates the nuclei. We use an asterisk (*) to designate antibonding orbitals such as this sigma antibonding orbital, σ^*.

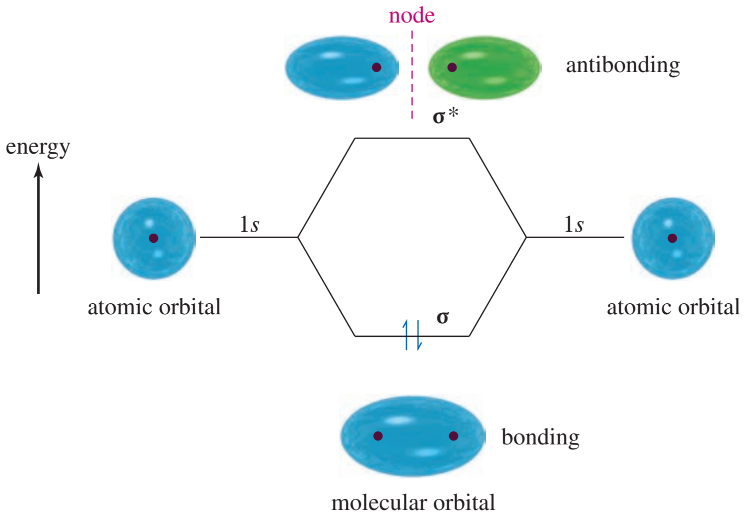


FIGURE 1-16 Relative energies of atomic and molecular orbitals. When the two hydrogen 1*s* orbitals overlap, a sigma bonding MO and a sigma antibonding MO result. The bonding MO is lower in energy than the atomic 1*s* orbital, and the antibonding orbital is higher in energy. Two electrons (represented by arrows) go into the bonding MO with opposite spins, forming a stable H_2 molecule. The antibonding orbital is vacant.

usually indicates that the orbital is antibonding. The antibonding MO is designated σ^* to indicate an antibonding (*), cylindrically symmetrical (σ) molecular orbital.

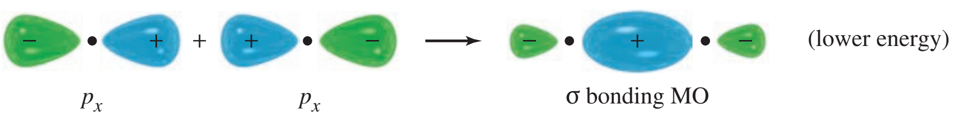
Figure 1-16 shows the relative energies of the atomic orbitals and the molecular orbitals of the H_2 system. When the 1*s* orbitals are in phase, the resulting molecular orbital is a σ bonding MO, with lower energy than that of a 1*s* atomic orbital. When two 1*s* orbitals overlap out of phase, they form an antibonding (σ^*) orbital with higher energy than that of a 1*s* atomic orbital. The two electrons in the H_2 system are found with paired spins in the sigma bonding MO, giving a stable H_2 molecule. Both bonding and antibonding orbitals exist in all molecules, but the antibonding orbitals (such as σ^*) are usually vacant in stable molecules. Antibonding molecular orbitals often participate in reactions, however.

PROBLEM-SOLVING HINT

In stable compounds, all or most of the bonding orbitals will be filled, and all or most of the antibonding orbitals will be empty.

1-13B Sigma Overlap Involving *p* Orbitals

When two *p* orbitals overlap along the line between the nuclei, a bonding orbital and an antibonding orbital result. Once again, most of the electron density is centered along the line between the nuclei. This linear overlap is another type of sigma bonding MO. The constructive overlap of two *p* orbitals along the line joining the nuclei forms a σ bond represented as follows:

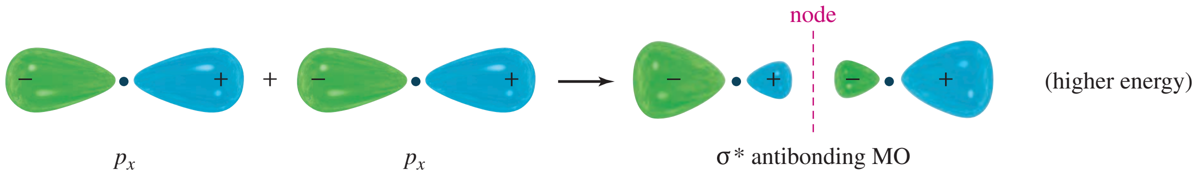


SOLVED PROBLEM 1-6

Draw the σ^* antibonding orbital that results from the destructive overlap of the two p_x orbitals just shown.

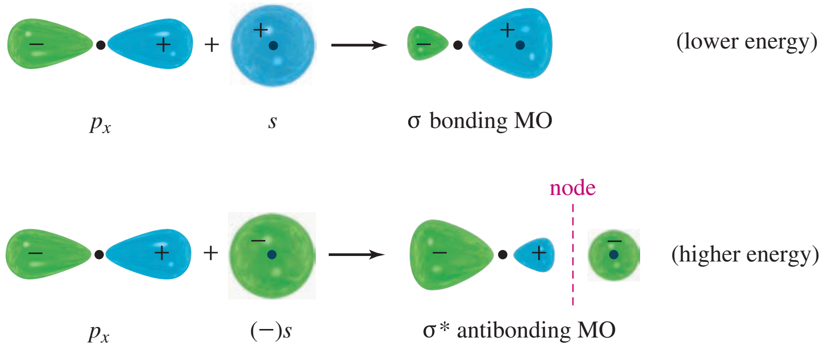
SOLUTION

This orbital results from the destructive overlap of lobes of the two *p* orbitals with opposite phases. If the signs are reversed on one of the orbitals, adding the two orbitals gives an antibonding orbital with a node separating the two nuclei:



(*continued*)

Overlap of an s orbital with a p orbital also gives a bonding MO and an antibonding MO, as shown in the following illustration. Constructive overlap of the s orbital with the p_x orbital gives a sigma bonding MO with its electron density centered along the line between the nuclei. Destructive overlap gives a σ^* antibonding orbital with a node separating the nuclei.



1-14 Pi Bonding

A **pi bond (π bond)** results from overlap between two p orbitals oriented perpendicular to the line connecting the nuclei (Figure 1-17). These parallel orbitals overlap sideways, with most of the electron density centered *above and below* the line connecting the nuclei. This overlap is parallel, not linear (a sigma bond is linear), so a pi molecular orbital is *not* cylindrically symmetrical. Figure 1-17 shows a π bonding MO and the corresponding π^* antibonding MO.

1-14A Single and Double Bonds

A **double bond** requires the presence of four electrons in the bonding region between the nuclei. The first pair of electrons goes into the sigma bonding MO, forming a strong sigma bond. The second pair of electrons cannot go into the same orbital or the same space. It goes into a pi bonding MO, with its electron density centered above and below the sigma bond.

This combination of one sigma bond and one pi bond is the normal structure of a double bond. Figure 1-18 shows the structure of ethylene, an organic molecule containing a carbon–carbon double bond.

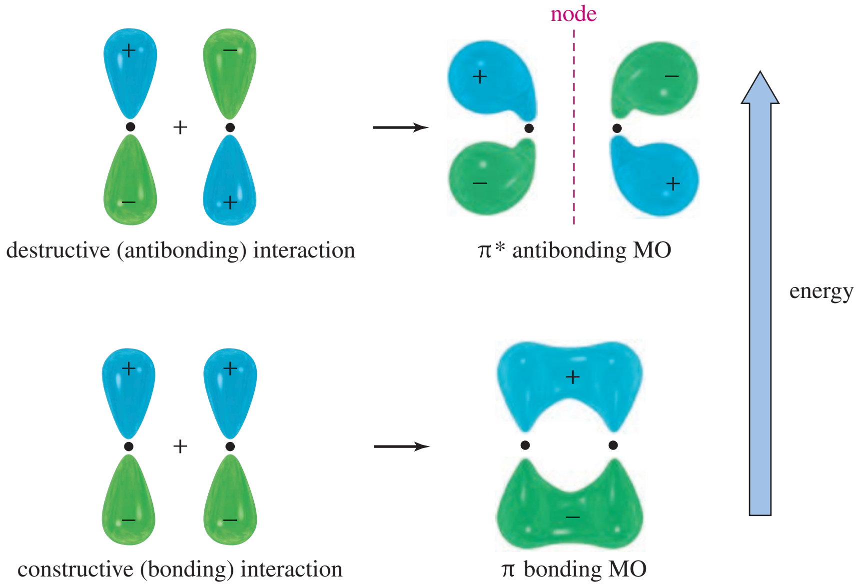


FIGURE 1-17 Pi bonding and antibonding molecular orbitals. The sideways overlap of two p orbitals leads to a π bonding MO and a π^* antibonding MO. A pi bond is not as strong as most sigma bonds.

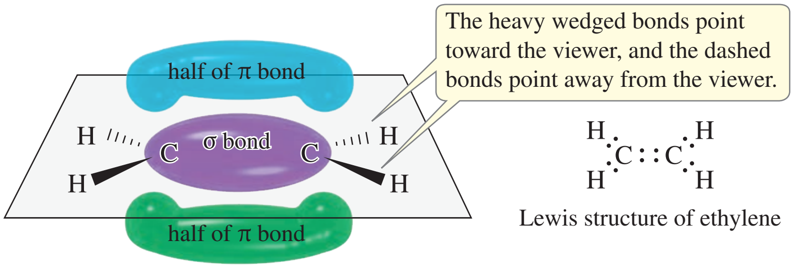


FIGURE 1-18 Structure of the double bond in ethylene. The first pair of electrons forms a σ bond. The second pair forms a π bond. The π bond has its electron density centered in two lobes above and below the σ bond. Together, the two lobes of the π bonding molecular orbital constitute one bond.

1-15 Hybridization and Molecular Shapes

Thus far, we have discussed bonds involving overlap of simple *s* and *p* atomic orbitals. Although these simple bonds are occasionally seen in organic compounds, they are not as common as bonds formed using **hybrid atomic orbitals.** Hybrid atomic orbitals result from the mixing of orbitals on the *same* atom. The geometry of these hybrid orbitals helps us to account for the actual structures and bond angles observed in organic compounds.

If we predict the bond angles of organic molecules using just the simple *s* and *p* orbitals, we expect bond angles of about 90°. The *s* orbitals are nondirectional, and the *p* orbitals are oriented at 90° to one another (see Figure 1-5). Experimental evidence shows, however, that bond angles in organic compounds are usually close to 109°, 120°, or 180° (Figure 1-19). A common way of accounting for these bond angles is the **valence-shell electron-pair repulsion theory (VSEPR theory):** Electron pairs repel each other, and the bonds and lone pairs around a central atom generally are separated by the largest possible angles. An angle of 109.5° is the largest possible separation for four pairs of electrons; 120° is the largest separation for three pairs; and 180° is the largest separation for two pairs. All the structures in Figure 1-19 have bond angles that separate their bonds about as far apart as possible.

The shapes of these molecules cannot result from bonding between simple *s* and *p* atomic orbitals. Although *s* and *p* orbitals have the lowest energies for isolated atoms in space, they are not the best for forming bonds. To explain the shapes of common organic molecules, we conclude that the *s* and *p* orbitals combine to form hybrid atomic orbitals that separate the electron pairs more widely in space and place more electron density in the bonding region between the nuclei.

1-15A *sp* Hybrid Orbitals

Orbitals can interact to form new orbitals. We have used this principle to form molecular orbitals by adding and subtracting atomic orbitals on *different* atoms. We can also add and subtract orbitals on the *same* atom. Consider the result, shown in Figure 1-20, when we combine a *p* orbital and an *s* orbital on the same atom.

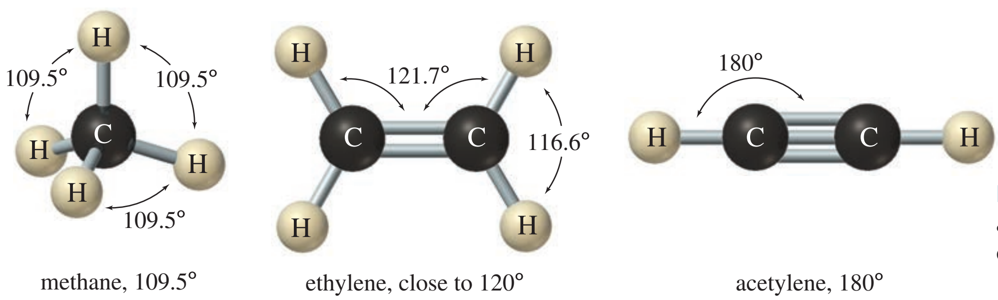


FIGURE 1-19 Common bond angles. Bond angles in organic compounds are usually close to 109°, 120°, or 180°.

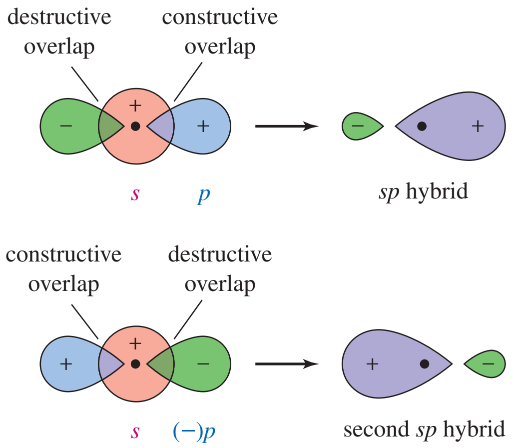


FIGURE 1-20 Formation of a pair of *sp* hybrid atomic orbitals. Addition of an *s* orbital to a *p* orbital gives an *sp* hybrid atomic orbital, with most of its electron density on one side of the nucleus. Adding the *p* orbital with opposite phase gives the other *sp* hybrid orbital, with most of its electron density on the opposite side of the nucleus from the first hybrid.

The resulting orbital is called an ***sp* hybrid orbital.** Its electron density is concentrated toward one side of the atom. We started with two orbitals (*s* and *p*), so we must finish with two *sp* hybrid orbitals. The second *sp* hybrid orbital results if we add the *p* orbital with the opposite phase (Figure 1-20).

The result of this hybridization is a pair of directional *sp* hybrid orbitals pointed in opposite directions. These hybridized orbitals provide enhanced electron density in the bonding region for a sigma bond toward the left of the atom and for another sigma bond toward the right. They give a bond angle of 180°, separating the bonding electrons as much as possible. In general, *sp* hybridization results in this **linear** bonding arrangement.

SOLVED PROBLEM 1-7

Draw the Lewis structure for beryllium hydride, BeH_2. Draw the orbitals that overlap in the bonding of BeH_2, and label the hybridization of each orbital. Predict the H—Be—H bond angle.

SOLUTION

First, draw a Lewis structure for BeH_2.

H:Be:H

There are only four valence electrons in BeH_2 (two from Be and one from each H), so the Be atom cannot have an octet. The bonding must involve orbitals on Be that give the strongest bonds (the most electron density in the bonding region) and also allow the two pairs of electrons to be separated as far as possible.

Hybrid orbitals concentrate the electron density in the bonding region, and *sp* hybrids give 180° separation for two pairs of electrons. Hydrogen cannot use hybridized orbitals, because the closest available *p* orbitals are the 2*p*'s, and they are much higher in energy than the 1*s*. The bonding in BeH_2 results from overlap of *sp* hybrid orbitals on Be with the 1*s* orbitals on hydrogen. Figure 1-21 shows how this occurs.

Application: Biology

Methanotrophs are bacteria or archaea that use methane as their source of carbon and energy. Those that live in the air use oxygen to oxidize methane to formaldehyde ($H_2C{=}O$) and CO_2. Those that live in anoxic marine sediments use sulfate (SO_4^{2-}) to oxidize methane to formaldehyde and CO_2, also reducing sulfate to H_2S.

1-15B sp^2 Hybrid Orbitals

For three bonds to be oriented as far apart as possible, bond angles of 120° are required. When an *s* orbital combines with two *p* orbitals, the resulting three hybrid orbitals are oriented at 120° angles to each other (Figure 1-22). These orbitals are called **sp^2 hybrid orbitals** because they are composed of one *s* and two *p* orbitals. The 120° arrangement is called **trigonal** geometry, in contrast to the linear geometry associated with *sp* hybrid orbitals. There remains an unhybridized *p* orbital (p_z) perpendicular to the plane of the three sp^2 hybrid orbitals.

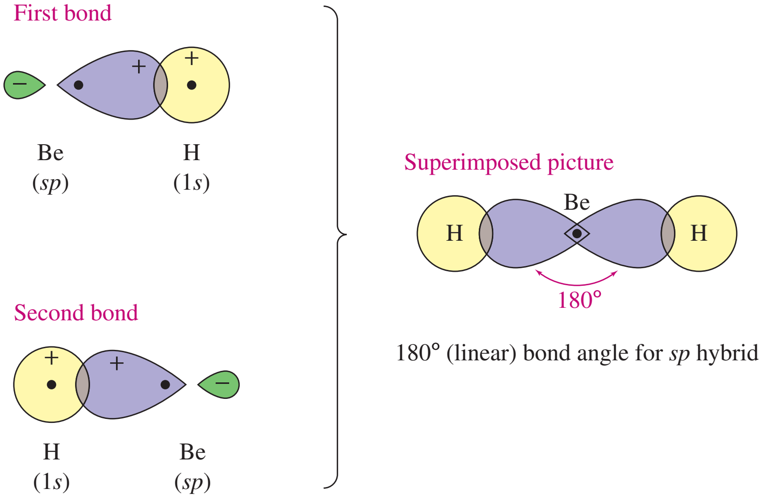


FIGURE 1-21 Linear geometry in the bonding of BeH_2. To form two sigma bonds, the two *sp* hybrid atomic orbitals on Be overlap with the 1*s* orbitals of hydrogen. The bond angle is 180° (linear).

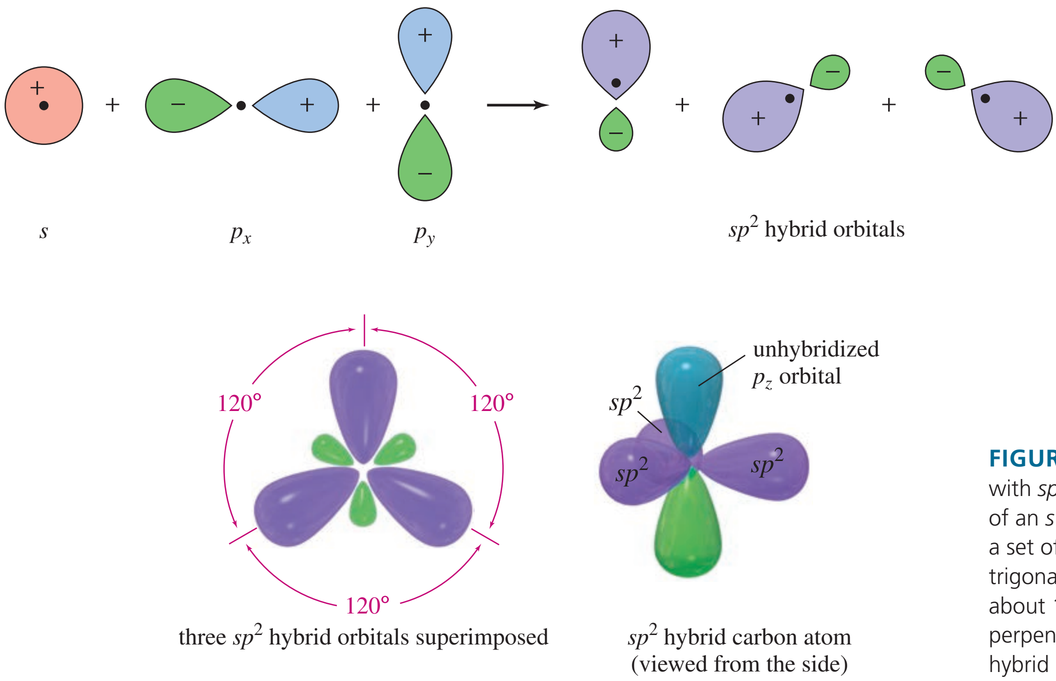


FIGURE 1-22 Trigonal geometry with sp^2 hybrid orbitals. Hybridization of an *s* orbital with two *p* orbitals gives a set of three sp^2 hybrid orbitals. This trigonal structure has bond angles of about 120°. The remaining *p* orbital is perpendicular to the plane of the three hybrid orbitals.

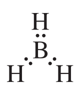
SOLVED PROBLEM 1-8

Borane (BH_3) is unstable under normal conditions, but it has been detected at low pressure.

(a) Draw the Lewis structure for borane.
(b) Draw a diagram of the bonding in BH_3, and label the hybridization of each orbital.
(c) Predict the H—B—H bond angle.

SOLUTION

There are only six valence electrons in borane, so the boron atom cannot have an octet. Boron has a single bond to each of the three hydrogen atoms.



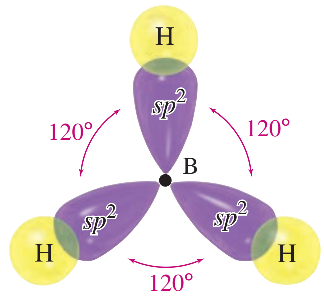
(continued)

PROBLEM-SOLVING HINT

The number of hybrid orbitals formed is always the same as the total number of *s* and *p* orbitals hybridized.

Number of Orbitals	Hybrid	Angle
2	*sp*	180°
3	sp^2	120°
4	sp^3	109.5°

The best bonding orbitals are those that provide the greatest electron density in the bonding region while keeping the three pairs of bonding electrons as far apart as possible. Hybridization of an *s* orbital with two *p* orbitals gives three sp^2 hybrid orbitals directed 120° apart. Overlap of these orbitals with the hydrogen 1*s* orbitals gives a planar, trigonal molecule. (Note that the small back lobes of the hybrid orbitals have been omitted.)



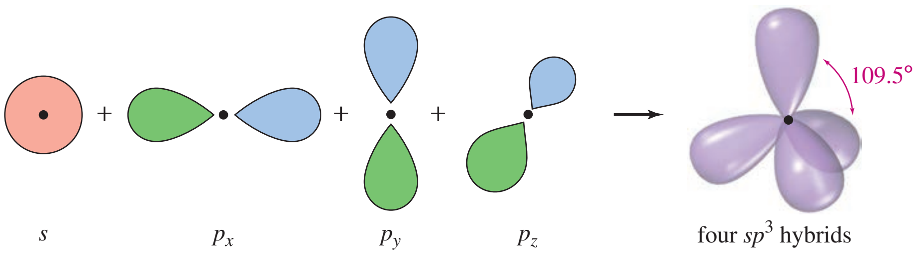


FIGURE 1-23 Tetrahedral geometry with sp^3 hybrid orbitals. Hybridization of an *s* orbital with all three *p* orbitals gives four sp^3 hybrid orbitals with tetrahedral geometry corresponding to 109.5° bond angles.

1-15C sp^3 Hybrid Orbitals

Many organic compounds contain carbon atoms that are bonded to four other atoms. When four bonds are oriented as far apart as possible, they form a regular tetrahedron (109.5° bond angles), as pictured in Figure 1-23. This **tetrahedral** arrangement can be explained by combining the *s* orbital with all three *p* orbitals. The resulting four orbitals are called **sp^3 hybrid orbitals** because they are composed of one *s* and three *p* orbitals.

Methane (CH_4) is the simplest example of sp^3 hybridization (Figure 1-24). The Lewis structure for methane has eight valence electrons (four from carbon and one from each hydrogen), corresponding to four C—H single bonds. Tetrahedral geometry separates these bonds by the largest possible angle, 109.5°.

1-16 Drawing Three-Dimensional Molecules

Figures 1-23 and 1-24 are more difficult to draw than the earlier figures because they depict three-dimensional objects on a two-dimensional surface. The p_z orbital should look like it points in and out of the page, and the tetrahedron should look three-dimensional. These drawings use perspective and the viewer's imagination to add the third dimension.

The use of perspective is difficult when a molecule is large and complicated. Organic chemists have developed a shorthand notation to simplify three-dimensional drawings. Dashed lines indicate bonds that go backward, away from the reader. Heavy wedge-shaped lines depict bonds that come forward, toward the reader. Straight lines are bonds in the plane of the page. Dashed lines and wedges show perspective in the second drawing of methane in Figure 1-24.

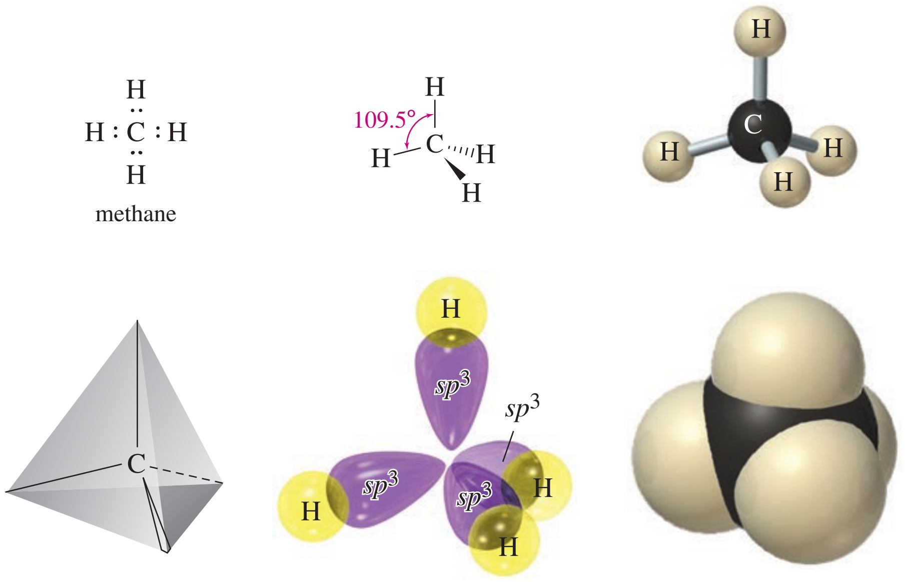
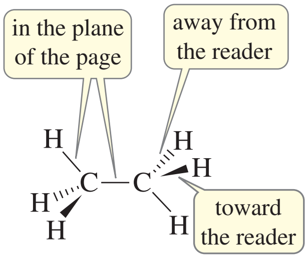
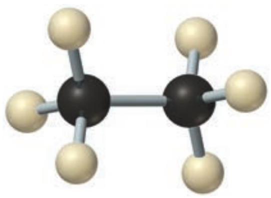


FIGURE 1-24 Several views of methane. Methane has tetrahedral geometry, using four sp^3 hybrid orbitals to form sigma bonds to the four hydrogen atoms.

The three-dimensional structure of ethane, C_2H_6, has the shape of two tetrahedra joined together. Each carbon atom is sp^3 hybridized, with four sigma bonds formed by the four sp^3 hybrid orbitals. Dashed lines represent bonds that go away from the viewer, wedges represent bonds that come out toward the viewer, and other bond lines are in the plane of the page. All the bond angles are close to 109.5°.



ethane

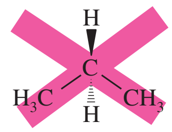


ethane

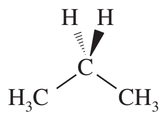
ethane

PROBLEM-SOLVING HINT

When showing perspective, do not draw another bond between the two bonds in the plane of the paper. Such a drawing shows an incorrect shape.



incorrect



correct

correct

PROBLEM 1-15

(a) Use your molecular models to make ethane, and compare the model with the preceding structures.
(b) Make a model of butane (C_4H_{10}), and draw this model using dashed lines and wedges to represent bonds going back and coming forward.

1-17 General Rules of Hybridization and Geometry

At this point, we can consider some general rules for determining the hybridization of orbitals and the bond angles of atoms in organic molecules. After stating these rules, we solve some problems to show how to use these rules.

Rule 1: Both sigma bonding electrons and lone pairs can occupy hybrid orbitals. The number of hybrid orbitals on an atom is computed by adding the number of sigma bonds and the number of lone pairs of electrons on that atom.

SUMMARY Hybridization and Geometry

Hybrid Orbitals	Hybridization	Geometry	Approximate Bond Angles
2	$s + p = sp$	linear	180°
3	$s + p + p = sp^2$	trigonal	120°
4	$s + p + p + p = sp^3$	tetrahedral	109.5°

Because the first bond to another atom is always a sigma bond, the number of hybrid orbitals may be computed by adding the number of lone pairs to the number of atoms bonded to the central atom.

> Rule 2: Use the hybridization and geometry that give the widest possible separation of the calculated number of bonds and lone pairs.

The number of hybrid orbitals obtained equals the number of atomic orbitals combined. Lone pairs of electrons take up more space than bonding pairs of electrons; thus, they compress the bond angles.

> Rule 3: If two or three pairs of electrons form a multiple bond between two atoms, the first bond is a sigma bond formed by a hybrid orbital. The second bond is a pi bond, consisting of two lobes above and below the sigma bond, formed by two unhybridized *p* orbitals (see the structure of ethylene in Figure 1-25). The third bond of a triple bond is another pi bond, perpendicular to the first pi bond (shown in Figure 1-26).

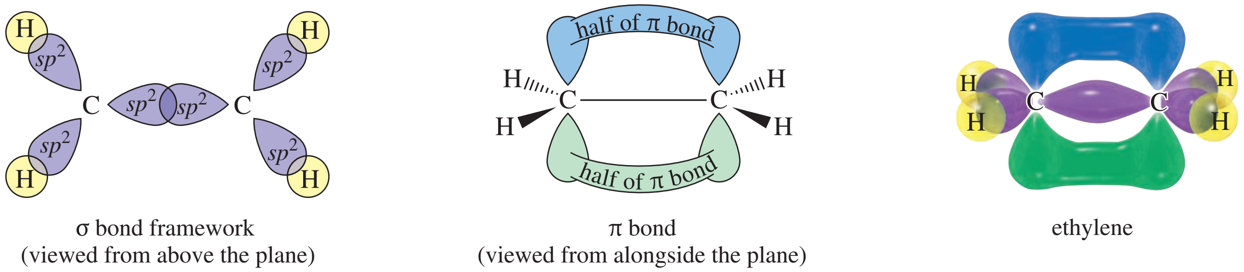


FIGURE 1-25 Planar geometry of ethylene. The carbon atoms in ethylene are sp^2 hybridized, with trigonal bond angles of about 120°. All the carbon and hydrogen atoms lie in the same plane.

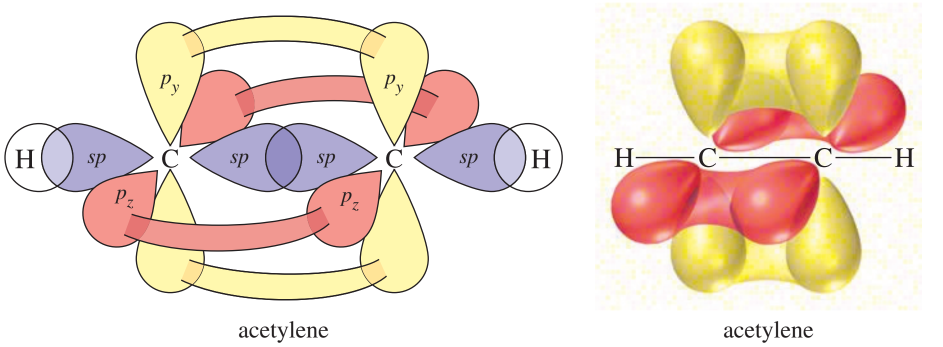


FIGURE 1-26 Linear geometry of acetylene. The carbon atoms in acetylene are *sp* hybridized, with linear (180°) bond angles. The triple bond contains one sigma bond and two perpendicular pi bonds.

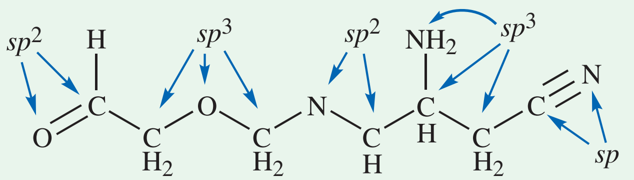
FOCUS Determining Hybridization of C, N, and O

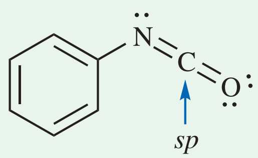
The most abundant elements in organic compounds are C, H, N, and O. H is not hybridized. When C, N, and O are in stable compounds with full octets, we can simplify the task of determining hybridization of these elements.

How to determine the hybridization of an atom? **By the number of pi bonds it has.**

If an atom (C, N, or O) has		Hybridization	Geometry
no pi bonds ⇨	no *p* orbital in pi bonding ⇨	***sp***^{3} hybridization ⇨	tetrahedral (109.5° angles)
1 pi bond ⇨	1 *p* orbital in pi bonding ⇨	***sp***^{2} hybridization ⇨	trigonal planar (120° angles)
2 pi bonds ⇨	2 *p* orbitals in pi bonding ⇨	*sp* hybridization ⇨	linear (180° angles)

Examples: Assign hydridization of each hybridized atom in these molecules.





All hybridized atoms in this structure are ***sp***^{2} except the carbon labeled *sp* (two pi bonds, requires two *p* orbitals). The N has a lone pair of electrons occupying one of its ***sp***^{2}-hybridized orbitals. The O has two lone pairs in two ***sp***^{2}-hybridized orbitals.

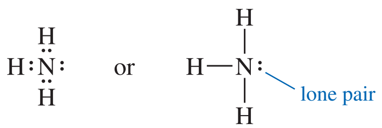
Solved Problems 1-9 through 1-13 show how to use these rules to predict the hybridization and bond angles in organic compounds.

SOLVED PROBLEM 1-9

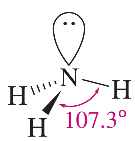
Predict the hybridization of the nitrogen atom in ammonia, NH_3. Draw a picture of the three-dimensional structure of ammonia, and predict the bond angles.

SOLUTION

The hybridization depends on the number of sigma bonds plus lone pairs. A Lewis structure provides this information.



In this structure, there are three sigma bonds and one pair of nonbonding electrons. Four hybrid orbitals are required, implying sp^3 hybridization and tetrahedral geometry around the nitrogen atom, with bond angles of about 109.5°. The resulting structure is much like that of methane, except that one of the sp^3 hybrid orbitals is occupied by a lone pair of electrons.



(*continued*)

The bond angles in ammonia (107.3°) are slightly smaller than the ideal tetrahedral angle, 109.5°. The nonbonding electrons are spread out more than a bonding pair of electrons, so they take up more space. The lone pair repels the electrons in the N—H bonds, compressing the bond angle.

PROBLEM 1-16

(a) Predict the hybridization of the oxygen atom in water, H_2O. Draw a picture of its three-dimensional structure, and explain why its bond angle is 104.5°.

(b) The electrostatic potential maps for ammonia and water are shown here. The structure of ammonia is shown within its EPM. Note how the lone pair creates a region of high electron potential (red), and the hydrogens are in regions of low electron potential (blue). Show how your three-dimensional structure of water corresponds with its EPM.

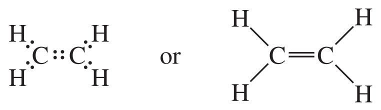
NH_3 H_2O

SOLVED PROBLEM 1-10

Predict the hybridization, geometry, and bond angles for ethylene (C_2H_4).

SOLUTION

The Lewis structure of ethylene is



Each carbon atom has an octet, and there is a double bond between the carbon atoms. Each carbon is bonded to three other atoms (three sigma bonds), and there are no lone pairs. The carbon atoms are sp^2 hybridized, and the bond angles are trigonal: about 120°. The double bond is composed of a sigma bond formed by overlap of two sp^2 hybridized orbitals, plus a pi bond formed by overlap of the unhybridized p orbitals remaining on the carbon atoms. Because the pi bond requires parallel alignment of its two p orbitals, the ethylene molecule must be planar (Figure 1-25).

PROBLEM 1-17

Predict the hybridization, geometry, and bond angles for the central atoms in
(a) but-2-ene, $CH_3CH{=}CHCH_3$. **(b)** $CH_3CH{=}NH$.

SOLVED PROBLEM 1-11

Predict the hybridization, geometry, and bond angles for the carbon atoms in acetylene, C_2H_2.

SOLUTION

The Lewis structure of acetylene is

H:C:::C:H or H—C≡C—H

Both carbon atoms have octets, but each carbon is bonded to just two other atoms, requiring two sigma bonds. There are no lone pairs. Each carbon atom is sp hybridized and linear (180° bond angles). The sp hybrid orbitals are generated from the s orbital and the p_x orbital (the p orbital directed along the line joining the nuclei). The p_y orbitals and the p_z orbitals are unhybridized.

PROBLEM-SOLVING HINT

Begin with a valid Lewis structure, and use hybrid orbitals for the sigma bonds and lone pairs. Use pi bonds between unhybridized p orbitals for the second and third bonds of double and triple bonds.

The **triple bond** is composed of one sigma bond, formed by overlap of *sp* hybrid orbitals, plus two pi bonds. One pi bond results from sideways overlap of the two p_y orbitals and another from sideways overlap of the two p_z orbitals (Figure 1-26).

PROBLEM 1-18

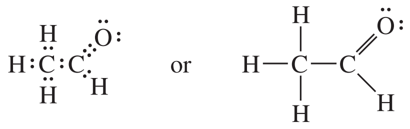
Predict the hybridization, geometry, and bond angles for the carbon and nitrogen atoms in acetonitrile ($CH_3{-}C{\equiv}N$:).

SOLVED PROBLEM 1-12

Predict the hybridization, geometry, and bond angles for the carbon and oxygen atoms in acetaldehyde (CH_3CHO).

SOLUTION

The Lewis structure for acetaldehyde is



The oxygen atom and both carbon atoms have octets. The CH_3 carbon atom is sigma bonded to four atoms, so it is sp^3 hybridized (and tetrahedral). The C$=$O carbon is bonded to three atoms (no lone pairs), so it is sp^2 hybridized, and its bond angles are about 120°.

We predict that the oxygen atom is sp^2 hybridized because it is bonded to one atom (carbon) and has two lone pairs, requiring a total of three hybrid orbitals. We cannot experimentally measure the angles of the lone pairs on oxygen, however, so it is impossible to confirm whether the oxygen atom is really sp^2 hybridized.

The double bond between carbon and oxygen looks just like the double bond in ethylene. There is a sigma bond formed by overlap of sp^2 hybrid orbitals and a pi bond formed by overlap of the unhybridized *p* orbitals on carbon and oxygen (Figure 1-27).

PROBLEM 1-19

1. Draw a Lewis structure for each compound.
2. Label the hybridization, geometry, and bond angles around each atom other than hydrogen.
3. Draw a three-dimensional representation (using wedges and dashed lines) of the structure.

(a) CO_2 **(b)** CH_3OCH_3 **(c)** $(CH_3)_3O^+$ **(d)** CH_3COOH
(e) CH_3CCH **(f)** CH_3CHNCH_3 **(g)** H_2CCO

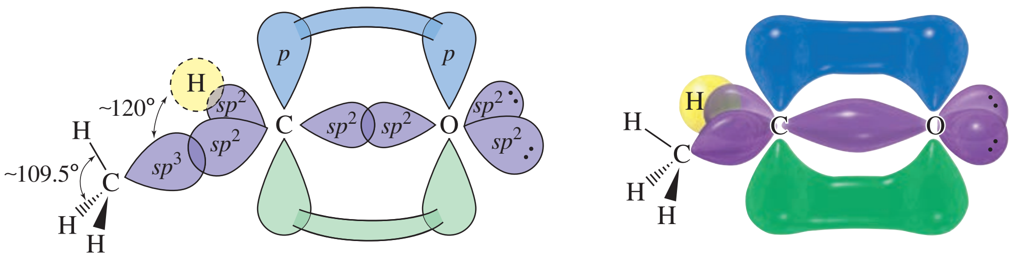
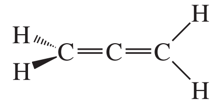


FIGURE 1-27 Structure of acetaldehyde. The CH_3 carbon in acetaldehyde is sp^3 hybridized, with tetrahedral bond angles of about 109.5°. The carbonyl (C$=$O) carbon is sp^2 hybridized, with bond angles of about 120°. The oxygen atom is probably sp^2 hybridized, but we cannot measure any bond angles to verify this prediction.

PROBLEM 1-20

Allene, $CH_2{=}C{=}CH_2$, has the structure shown below. Explain how the bonding in allene requires the two $=CH_2$ groups at its ends to be at right angles to each other.



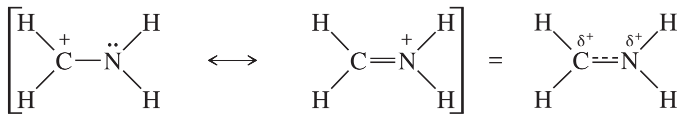
allene

SOLVED PROBLEM 1-13

In Sections 1-7 and 1-9, we considered the electronic structure of $[CH_2NH_2]^+$. Predict its hybridization, geometry, and bond angles.

SOLUTION

This is a tricky question. This ion has two important resonance forms:



resonance forms

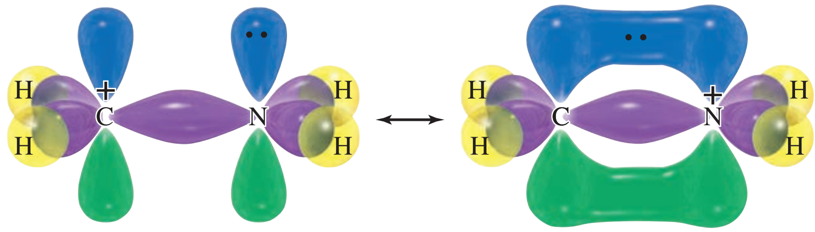
combined representation

When resonance is involved, different resonance forms may suggest different hybridization and bond angles, but a real molecule can have only one set of bond angles, which must be compatible with all the important resonance forms. The bond angles imply the hybridization of the atoms, which also must be the same in all the resonance forms.

Looking at either resonance form for $[CH_2NH_2]^+$, we would predict sp^2 hybridization (120° bond angles) for the carbon atom; however, the first resonance form suggests sp^3 hybridization for nitrogen (109° bond angles), and the second suggests sp^2 hybridization (120° bond angles). Which is correct?

Experiments show that the bond angles on both carbon and nitrogen are about 120°, implying sp^2 hybridization. This nitrogen cannot be sp^3 hybridized because there must be an unhybridized p orbital available to form the pi bond in the second resonance form. In the first resonance form, we picture the lone pair residing in this unhybridized p orbital.

In general, resonance-stabilized structures have bond angles appropriate for the largest number of pi bonds needed at each atom—that is, with unhybridized p orbitals available for all the pi bonds shown in any important resonance form.



PROBLEM-SOLVING HINT

To predict the hybridization and geometry of an atom in a resonance hybrid, consider the resonance form with the most pi bonds to that atom. An atom involved in resonance generally will not be sp^3 hybridized because it needs at least one unhybridized p orbital for pi-bonding overlap.

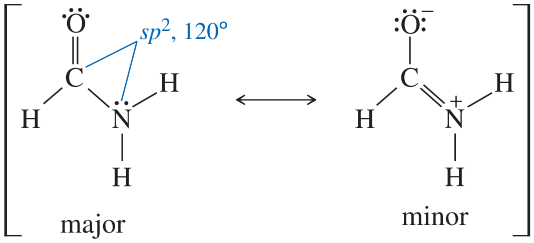
PROBLEM 1-21 (PARTIALLY SOLVED)

1. Draw the important resonance forms for each compound.
2. Label the hybridization and bond angles around each atom other than hydrogen.
3. Use a three-dimensional drawing to show where the electrons are pictured to be in each resonance form.

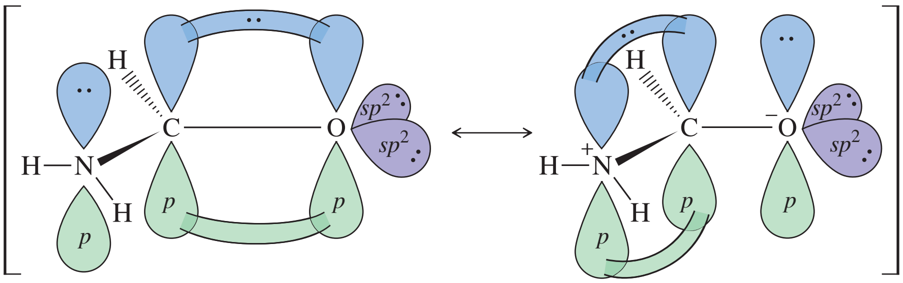
(a) $HCONH_2$

SOLUTION

This compound has a carbonyl (C=O) group that is not obvious in the condensed formula. An important resonance form delocalizes the nonbonding electrons on nitrogen into a pi bond with carbon. This pi bonding requires the overlap of unhybridized *p* orbitals, requiring sp^2 hybridization on both carbon and nitrogen, and 120° bond angles. Then O is probably sp^2 hybridized, but we cannot confirm that assumption because there are no bond angles on O.



resonance forms



(b) $[CH_2OH]^+$ **(c)** $[CH_2CHO]^-$ **(d)** $[CH_3CHNO_2]^-$
(e) $[CH_2CN]^-$ **(f)** $B(OH)_3$ **(g)** ozone (O_3, bonded OOO)

1-18 Bond Rotation

Some bonds rotate easily, but others do not. When we look at a structure, we must recognize which bonds rotate and which do not. If a bond rotates easily, each molecule can rotate through the different angular arrangements of atoms. If a bond cannot rotate, however, different angular arrangements may be distinct compounds with different properties.

1-18A Rotation of Single Bonds

In ethane ($CH_3—CH_3$), both carbon atoms are sp^3 hybridized and tetrahedral. Ethane looks like two methane molecules that have each had a hydrogen plucked off (to form a methyl group) and are joined by overlap of their sp^3 orbitals (Figure 1-28).

We can draw many structures for ethane, differing only in how one methyl group is twisted in relation to the other one. Such structures, differing only in rotations about a single bond, are called *conformations*. Two of the infinite number of conformations of ethane are shown in Figure 1-28. Construct a molecular model of ethane, and twist the model into these two conformations.

Which of these structures in Figure 1-28 for ethane is the “right” one? Are the two methyl groups lined up so that their C—H bonds are parallel (*eclipsed*), or are they *staggered*, as in the drawing on the right? The answer is that both structures, and all the possible structures in between, are correct structures for ethane, and a real ethane molecule rotates through all these conformations. The two carbon atoms are bonded by overlap of their sp^3 orbitals to form a sigma bond along the line between the carbons. The magnitude of this sp^3–sp^3 overlap remains nearly the same during rotation because the sigma bond is cylindrically symmetrical about the line joining the carbon nuclei. No matter how you turn one of the methyl groups, its sp^3 orbital still overlaps with the sp^3 orbital of the other carbon atom.

PROBLEM-SOLVING HINT

Drawings that differ only by rotations of single bonds usually represent the same compound. For example, the following drawings all represent *n*-butane:

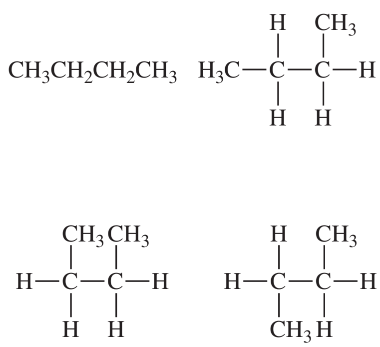
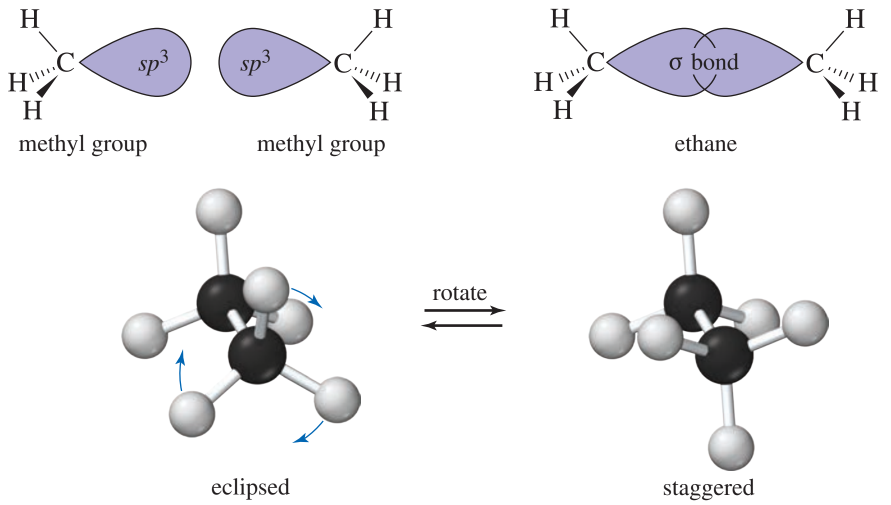


FIGURE 1-28 Rotation of single bonds. Ethane is composed of two methyl groups bonded by overlap of their sp^3 hybrid orbitals. These methyl groups may rotate with respect to each other.



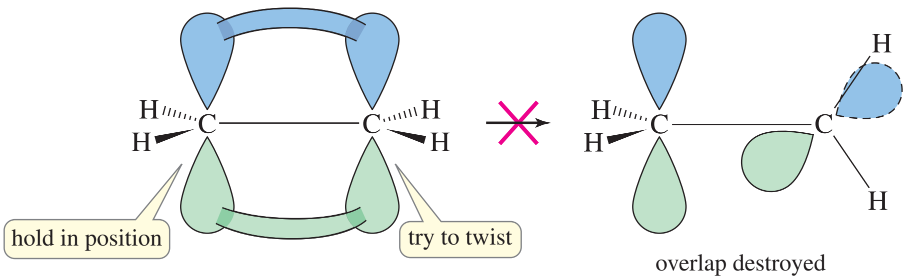


FIGURE 1-29 Rigidity of double bonds. Ethylene is not easily twisted because twisting destroys the overlap of its pi bond.

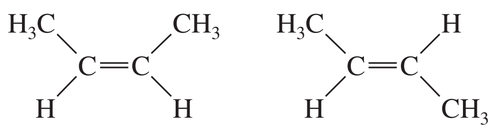
1-18B Rigidity of Double Bonds

Not all bonds allow free rotation; ethylene, for example, is quite rigid (Figure 1-29). In ethylene, the double bond between the two CH_2 groups consists of a sigma bond and a pi bond. When we twist one of the two CH_2 groups, the sigma bond is unaffected but the pi bond loses its overlap. The two *p* orbitals cannot overlap when the two ends of the molecule are at right angles, and the pi bond is effectively broken in this geometry.

We can make the following generalization:

> Rotation about single bonds is allowed, but double bonds are rigid and cannot be twisted under normal conditions.

Because double bonds are rigid, we can separate and isolate compounds that differ only in how their substituents are arranged on a double bond. For example, the double bond in but-2-ene ($CH_3—CH═CH—CH_3$) prevents the two ends of the molecule from rotating. Two different compounds are possible, and they have different physical properties:



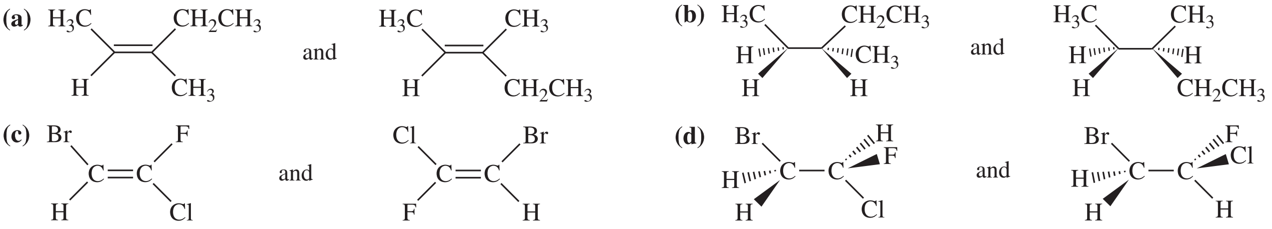
cis-but-2-ene
bp = 3.7 °C

trans-but-2-ene
bp = 0.9 °C

The molecule with the methyl groups on the same side of the double bond is called *cis*-but-2-ene, and the one with the methyl groups on opposite sides is called *trans*-but-2-ene. These kinds of molecules are discussed further in Section 1-19B.

PROBLEM 1-22

For each pair of structures, determine whether they represent different compounds or a single compound.



PROBLEM 1-23

Two compounds with the formula $CH_3—CH{=}N—CH_3$ are known.

(a) Draw a Lewis structure for this molecule, and label the hybridization of each carbon and nitrogen atom.

(b) What two compounds have this formula?

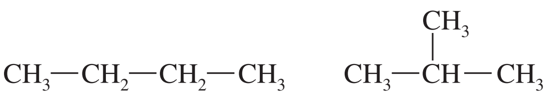
(c) Explain why only one compound with the formula $(CH_3)_2CNCH_3$ is known.

1-19 Isomerism

Isomers are different compounds with the same molecular formula. There are several types of isomerism in organic compounds, and we will cover them in detail in Chapter 5 (Stereochemistry). For now, we need to recognize the two large classes of isomers: constitutional isomers and stereoisomers.

1-19A Constitutional Isomerism

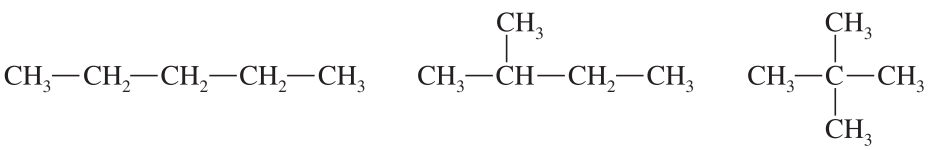
Constitutional isomers (or **structural isomers**) are isomers that differ in their bonding sequence; that is, their atoms are connected differently. Let's use butane as an example. If you were asked to draw a structural formula for C_4H_{10}, either of the following structures would be correct:



n-butane isobutane

These two compounds are isomers because they are different compounds with different properties, yet they have the same molecular formula. They are constitutional isomers because their atoms are connected differently. The first compound (*n*-butane for "normal" butane) has its carbon atoms in a straight chain four carbons long. The second compound ("isobutane" for "an isomer of butane") has a branched structure with a longest chain of three carbon atoms and a methyl side chain.

There are three constitutional isomers of pentane (C_5H_{12}), whose common names are *n*-pentane, isopentane, and neopentane. The number of isomers increases rapidly as the number of carbon atoms increases.



n-pentane isopentane neopentane

Constitutional isomers may differ in ways other than the branching of their carbon chain. They may differ in the position of a double bond or other group or

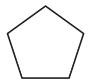
PROBLEM-SOLVING HINT

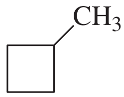
Constitutional isomers (structural isomers) differ in the order in which their atoms are bonded.

by having a ring or some other feature. Notice how the following constitutional isomers all differ by the ways in which atoms are bonded to other atoms. (Check the number of hydrogens bonded to each carbon.) These compounds are not isomers of the pentanes just shown, however, because these have a different molecular formula (C_5H_{10}).

$H_2C{=}CH{-}CH_2CH_2CH_3$
pent-1-ene

$CH_3{-}CH{=}CH{-}CH_2CH_3$
pent-2-ene


cyclopentane



methylcyclobutane

1-19B Stereoisomers

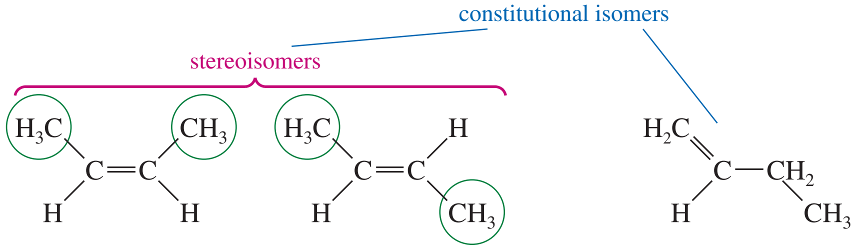
PROBLEM-SOLVING HINT

Stereoisomers are different compounds that differ only in how their atoms are oriented in space.

Stereoisomers are isomers that differ only in how their atoms are oriented in *sp*ace. Their atoms are bonded in the same order, however. For example, *cis*- and *trans*-but-2-ene have the same connections of bonds, so they are not constitutional isomers. They are stereoisomers because they differ only in the spatial orientation of the groups attached to the double bond. The cis isomer has the two methyl groups on the same side of the double bond, and the trans isomer has them on opposite sides. In contrast, but-1-ene is a constitutional isomer of *cis*- and *trans*-but-2-ene.

PROBLEM-SOLVING HINT

Similar groups on the same side of the double bond: *cis*. Similar groups on opposite sides of the double bond: *trans*.



cis-but-2-ene *trans*-but-2-ene but-1-ene

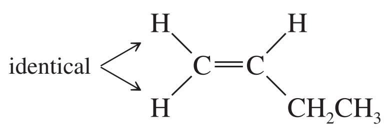
Cis and trans isomers are only one type of stereoisomerism. The study of the structure and chemistry of stereoisomers is called **stereochemistry.** We will encounter stereochemistry throughout our study of organic chemistry, and Chapter 5 is devoted entirely to this field.

Application: Drugs

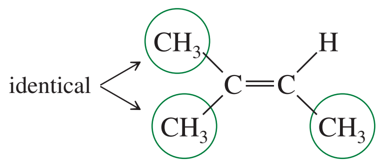
Stereoisomers can have different therapeutic effects. Quinine, a natural product isolated from the bark of the cinchona tree, was the first compound effective against malaria. Quinidine, a stereoisomer of quinine, is used to treat irregular heartbeat.

Cis-trans isomers are also called **geometric isomers** because they differ in the geometry of the groups on a double bond. The cis isomer is always the one with similar groups on the same side of the double bond, and the trans isomer has similar groups on opposite sides of the double bond.

To have cis-trans isomerism, there must be two *different* groups on each end of the double bond. For example, but-1-ene has two identical hydrogens on one end of the double bond. Reversing their positions does not give a different compound. Similarly, 2-methylbut-2-ene has two identical methyl groups on one end of the double bond. Reversing the methyl groups does not give a different compound. These compounds cannot show cis-trans isomerism.



but-1-ene
no cis or trans



2-methylbut-2-ene
no cis or trans

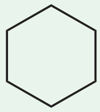
FOCUS Types of Isomers

Isomers are different compounds with the same molecular formula, so the first test for possible isomerism is whether compounds have the same molecular formula.

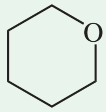
These compounds are different because they have different molecular formulas. They are not isomers; they are just different compounds.

$CH_3CH_2CH_2CH_2CH_3$
C_5H_{12}

C_6H_{14}



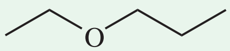
C_6H_{12}



$C_5H_{10}O$

The following compounds have the same molecular formula ($C_5H_{12}O$), and *they are the same compound*. They are drawn differently, but the bonding between atoms is identical.

$CH_3CH_2CH_2OCH_2CH_3$



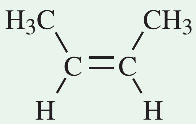
CH_3CH_2–O–$CH_2CH_2CH_3$

The following compounds have the same molecular formula ($C_5H_{12}O$), but they are different, structurally distinct compounds. They can be described in a way that distinguishes each from the others. These structures are **constitutional isomers** or **structural isomers**.

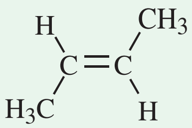
$CH_3CH_2CH_2OCH_2CH_3$

Stereoisomers are compounds with the same molecular formula and the same bonding between atoms, but they differ only by the positions of the atoms in space. The uses of models, and the ability to visualize molecules in three dimensions, are helpful for determining stereoisomers.

Of the several types of stereoisomers, only **geometric isomers** (**cis-trans isomers**) are introduced here. The example in the text shows *cis*- and *trans*-but-2-ene. *Cis* has the two $\mathbf{CH_3}$ groups on the same side of the double bond, and *trans* has them on opposite sides. They are stereoisomers: different compounds with the same molecular formula and same bonding sequence that differ only in the orientation of the groups.



cis-but-2-ene



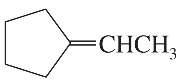
trans-but-2-ene

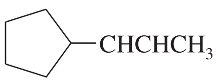
geometric (cis-trans) isomers, a type of stereoisomer

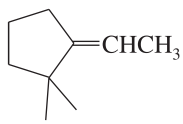
PROBLEM 1-24

Which of the following compounds show cis-trans isomerism? Draw the cis and trans isomers of those that do.

(a) $CHF{=}CHF$ **(b)** $F_2C{=}CH_2$ **(c)** $CH_2{=}CH{-}CH_2{-}CH_3$

(d) 

(e) 

(f) 

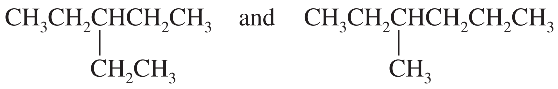
PROBLEM-SOLVING HINT

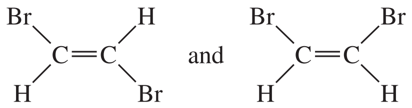
Two identical groups on one of the double-bonded carbons implies no cis-trans isomerism.

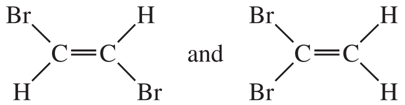
PROBLEM 1-25

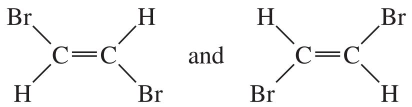
Give the relationship between the following pairs of structures. The possible relationships are:

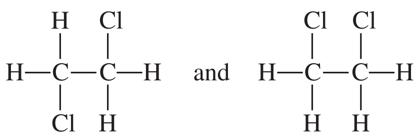
same compound — *constitutional isomers (structural isomers)*
cis-trans isomers — *not isomers (different molecular formula)*

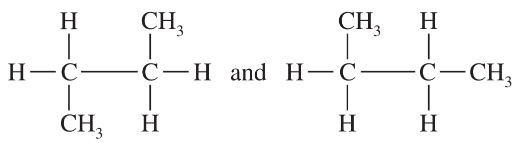
(a) 

(b) 

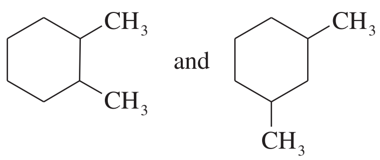
(c) 

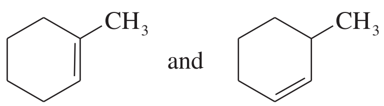
(d) 

(e) 

(f) 

(g) $CH_3-CH_2-CH_2-CH_3$ and $CH_3-CH=CH-CH_3$
(h) $CH_2=CH-CH_2CH_2CH_3$ and $CH_3-CH=CH-CH_2CH_3$
(i) $CH_2=CHCH_2CH_2CH_3$ and $CH_3CH_2CH_2CH=CH_2$

(j) 

(k) 

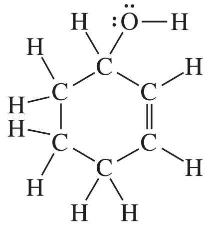
Essential Terms

Each chapter ends with a glossary that summarizes the most important new terms in the chapter. These glossaries are more than just a dictionary to look up unfamiliar terms as you encounter them (the index serves that purpose). The glossary is one of the tools for reviewing the chapter. You can read carefully through the glossary to see if you understand and remember all the terms and associated chemistry mentioned there. Anything that seems unfamiliar should be reviewed by turning to the page number given in the glossary listing.

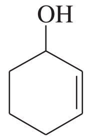
cis-trans isomers **(geometric isomers)** Stereoisomers that differ in their cis-trans arrangement on a double bond or on a ring. The cis isomer has similar groups on the same side, and the trans isomer has similar groups on opposite sides. (p. 82)

constitutional isomers **(structural isomers)** Isomers whose atoms are connected differently; they differ in their bonding sequence. (p. 81)

covalent bonding	Bonding that occurs by the sharing of electrons in the region between two nuclei. (p. 43)
single bond:	A covalent bond that involves the sharing of one pair of electrons. (p. 45)
double bond:	A covalent bond that involves the sharing of two pairs of electrons. (p. 45)
triple bond:	A covalent bond that involves the sharing of three pairs of electrons. (p. 45)
degenerate orbitals	Orbitals with identical energies. (p. 40)
delocalized charge	A charge that is spread out over two or more atoms. We usually draw resonance forms to show how the charge can appear on each of the atoms sharing the charge. (p. 50)
dipole moment (μ)	A measure of the polarity of a bond (or a molecule), proportional to the product of the charge separation times the bond length. (p. 46)
double bond	A bond containing four electrons between two nuclei. One pair of electrons forms a sigma bond, and the other pair forms a pi bond. (p. 68)
electron density	The relative probability of finding an electron in a certain region of space. (p. 39)
electronegativity	A measure of an element's ability to attract electrons. Elements with higher electronegativities attract electrons more strongly. (p. 46)
electrostatic potential map (EPM)	A computer-calculated molecular representation that uses colors to show the charge distribution in a molecule. In most cases, the EPM uses red to show electron-rich regions (most negative electrostatic potential) and blue or purple to show electron-poor regions (most positive electrostatic potential). The intermediate colors orange, yellow, and green show regions with intermediate electrostatic potentials. (p. 46)
empirical formula	The ratios of atoms in a compound. (p. 62) See also **molecular formula**.
formal charges	A method for keeping track of charges, showing what charge would be on an atom in a particular Lewis structure. (p. 47)
geometric isomers	(IUPAC term: **cis-trans isomers**) Stereoisomers that differ in their cis-trans arrangement on a double bond or on a ring. (p. 82)
Hund's rule	When there are two or more unfilled orbitals of the same energy (degenerate orbitals), the lowest-energy configuration places the electrons in different orbitals (with parallel spins) rather than pairing them in the same orbital. (p. 42)
hybrid atomic orbital	A directional orbital formed from a combination of *s* and *p* orbitals on the same atom. (pp. 65–68) ***sp* hybrid orbitals** give two orbitals with a bond angle of 180° (**linear** geometry). **sp^2 hybrid orbitals** give three orbitals with bond angles of 120° (**trigonal** geometry). **sp^3 hybrid orbitals** give four orbitals with bond angles of 109.5° (**tetrahedral** geometry).
ionic bonding	Bonding that occurs by the attraction of oppositely charged ions. Ionic bonding usually results in the formation of a large, three-dimensional crystal lattice. (p. 43)
isomers	Different compounds with the same molecular formula. (p. 81) **Constitutional isomers (structural isomers)** are connected differently; they differ in their bonding sequence. **Cis-trans isomers (geometric isomers)** are stereoisomers that differ in their cis-trans arrangement on a double bond or on a ring. **Stereoisomers** differ only in how their atoms are oriented in space. **Stereochemistry** is the study of the structure and chemistry of stereoisomers.
isotopes	Atoms with the same number of protons but different numbers of neutrons; atoms of the same element but with different atomic masses. (p. 39)
LCAO	**(linear combination of atomic orbitals)** Wave functions can add to each other to produce the wave functions of new orbitals. The number of new orbitals generated equals the original number of orbitals. (p. 65)
Lewis structure	A structural formula that shows all valence electrons, with the bonds symbolized by dashes (—) or by pairs of dots, and nonbonding electrons symbolized by dots. (p. 44)
line–angle formula	**(skeletal structure, stick figure)** A shorthand structural formula with bonds represented by lines. Carbon atoms are implied wherever two lines meet or a line begins or bends. Atoms other than C and H are drawn in, but hydrogen atoms are not shown unless they are on an atom that is drawn. Each carbon atom is assumed to have enough hydrogens to give it four bonds. (p. 60)



Lewis structure of cyclohex-2-en-1-ol



cyclohex-2-en-1-ol
equivalent line–angle formula

lone pair A pair of nonbonding electrons. (p. 44)

molecular formula The number of atoms of each element in one molecule of a compound. The **empirical formula** simply gives the ratios of atoms of the different elements. For example, the molecular formula of glucose is $C_6H_{12}O_6$. Its empirical formula is CH_2O. Neither the molecular formula nor the empirical formula gives structural information. (p. 62)

molecular orbital (MO) An orbital formed by the overlap of atomic orbitals on different atoms. MOs can be either bonding or antibonding, but only the bonding MOs are filled in most stable molecules. (pp. 65–68)

A **bonding molecular orbital** places a large amount of electron density in the bonding region between the nuclei. The energy of an electron in a bonding MO is lower than it is in an atomic orbital.

An **antibonding molecular orbital** places most of the electron density outside the bonding region. The energy of an electron in an antibonding MO is higher than it is in an atomic orbital.

node In an orbital, a region of space with zero electron density. (p. 40)

nodal plane In an orbital, a flat (planar) region of space with zero electron density. (p. 40)

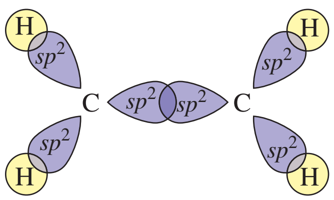
nonbonding electrons Valence electrons that are not used for bonding. A pair of nonbonding electrons is often called a **lone pair.** (p. 44)

octet rule Atoms generally form bonding arrangements that give them filled shells of electrons (noble-gas configurations). For the second-row elements, this configuration has eight valence electrons. (p. 43)

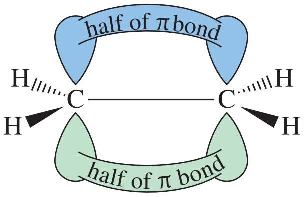
orbital An allowed energy state for an electron bound to a nucleus; the probability function that defines the distribution of electron density in space. The *Pauli exclusion principle* states that up to two electrons can occupy each orbital if their spins are paired. (p. 39)

organic chemistry New definition: The chemistry of carbon compounds. Old definition: The study of compounds derived from living organisms and their natural products. (p. 37)

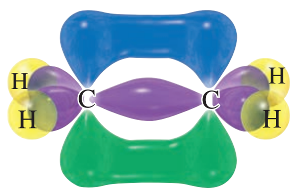
pi bond (π bond) A bond formed by sideways overlap of two *p* orbitals. A pi bond has its electron density in two lobes, one above and one below the line joining the nuclei. (p. 68)



σ bond framework
(viewed from above the plane)



π bond
(viewed from alongside the plane)



ethylene

polar covalent bond A covalent bond in which electrons are shared unequally. A bond with equal sharing of electrons is called a **nonpolar covalent bond.** (p. 46)

resonance hybrid A molecule or ion for which two or more valid Lewis structures can be drawn, differing only in the placement of the valence electrons. These Lewis structures are called **resonance forms** or **resonance structures**. Individual resonance forms do not exist, but we can estimate their relative energies. The more important (lower-energy) structures are called **major contributors**, and the less important (higher-energy) structures are called **minor contributors**. When a charge is spread over two or more atoms by resonance, it is said to be **delocalized** and the molecule is said to be **resonance stabilized**. (pp. 50–51)

sigma bond (σ bond) A bond with most of its electron density centered along the line joining the nuclei; a cylindrically symmetrical bond. Single bonds are normally sigma bonds. (p. 65)

stereochemistry The study of the structure and chemistry of stereoisomers. (p. 82)

stereoisomers Isomers that differ only in how their atoms are oriented in space. (p. 82)

structural formulas A **complete structural formula** (such as a Lewis structure) shows all the atoms and bonds in the molecule. A **condensed structural formula** shows each central atom along with the atoms bonded to it. A **line–angle formula** (sometimes called a **skeletal structure** or **stick figure**) assumes that there is a carbon atom wherever two lines meet or a line begins or ends. See Section 1-10 for examples. (pp. 58–61)

structural isomers (IUPAC term: **constitutional isomers**) Isomers whose atoms are connected differently; they differ in their bonding sequence. (p. 81)

triple bond A bond containing six electrons between two nuclei. One pair of electrons forms a sigma bond, and the other two pairs form two pi bonds at right angles to each other. (p. 77)

valence The number of bonds an atom usually forms. (p. 45)

valence electrons Those electrons that are in the outermost shell. (p. 42)

vitalism The belief that syntheses of organic compounds require the presence of a "vital force." (p. 37)

VSEPR theory **(valence-shell electron-pair repulsion theory)** Bonds and lone pairs around a central atom tend to be separated by the largest possible angles: about 180° for two, 120° for three, and 109.5° for four. (p. 69)

wave function (ψ) The mathematical description of an orbital. The square of the wave function (ψ^2) is proportional to the electron density. (p. 64)

Essential Problem-Solving Skills in Chapter 1

Each skill is followed by problem numbers exemplifying that particular skill.

1	Write the electronic configurations for the elements hydrogen through neon. Explain how electronic configurations determine the electronegativities and bonding properties of these elements, and how the third row elements (e.g., Si, P, and S) differ from them.	Problems 1-26, 27, 28, and 30
2	Draw all possible structures corresponding to a given molecular formula.	Problems 1-34, 35, and 36
3	Draw and interpret Lewis, condensed, and line–angle structural formulas. Calculate formal charges.	Problems 1-30, 31, 32, 33, 34, 37, 38, 41, 42, and 45
4	Predict patterns of covalent and ionic bonding involving C, H, O, N, and the halogens. Identify resonance-stabilized structures and compare the relative importance of their resonance forms.	Problems 1-26, 29, 36, 40, 41, 42, 43, 44, 45, and 46
	Problem-Solving Strategy: Drawing and Evaluating Resonance Forms	Problems 1-26, 41, 42, 43, 44, 45, 46, 53, and 54
5	Calculate empirical and molecular formulas from elemental composition.	Problems 1-47 and 48
6	Draw the structure of a single bond, a double bond, and a triple bond.	Problems 1-52, 55, 57, and 60
7	Predict the hybridization and geometry of the atoms in a molecule, and draw a three-dimensional representation of the molecule.	Problems 1-49, 51, 52, 53, 56, 57, and 60
8	Draw the resonance forms of a resonance hybrid, and predict its hybridization and geometry. Predict which resonance form is the major contributor.	Problems 1-43, 44, 52, 53, and 54
9	Identify constitutional isomers and stereoisomers, and predict which compounds can exist as constitutional isomers and as cis-trans (geometric) isomers.	Problems 1-49, 56, 58, and 59

Study Problems

It's easy to fool yourself into thinking you understand organic chemistry when you actually may not. As you read through this book, all the facts and ideas may make sense, yet you have not learned to combine and use those facts and ideas. An examination is a painful time to learn that you do not really understand the material.

The best way to learn organic chemistry is to use it. You will certainly need to read and reread all the material in the chapter, but this level of understanding is just the beginning. Problems are provided so that you can work with the ideas, applying them to new compounds and new reactions that you have never seen before. By working problems, you force yourself to use the material and fill in the gaps in your understanding. You also increase your level of self-confidence and your ability to do well on exams.

Several kinds of problems are included in each chapter. There are problems within the chapters, providing examples and drill for the material as it is covered. Work these problems as you read through the chapter to ensure your understanding as you go along. Answers to many of these in-chapter problems are found at the back of this book. Study Problems at the end of each chapter give you additional experience using the material, and they force you to think in depth about the ideas. Problems with red stars (*) are more difficult problems that require extra thought and perhaps some extension of the material presented in the chapter.

Taking organic chemistry without working the problems is like skydiving without a parachute. Initially there is a breezy sense of freedom and daring. But then, there is the inevitable jolt that comes at the end for those who went unprepared.

1-26 **(a)** Draw the resonance forms for CO_2 (bonded O—C—O).
(b) Draw the resonance forms for ozone (bonded O—O—O).
(c) Carbon dioxide has one more resonance form than ozone. Explain why this structure is not possible for ozone.

1-27 Name the element that corresponds to each electronic configuration.
(a) $1s^2 2s^2 2p^6 3s^1$ **(b)** $1s^2 2s^2 2p^6$ **(c)** $1s^2 2s^2 2p^1$ **(d)** $1s^2 2s^2 2p^6 3s^2$

1-28 There is a small portion of the periodic table that you must know to do organic chemistry. Construct this part from memory, using the following steps.
(a) From memory, make a list of the elements in the first two rows of the periodic table, together with their numbers of valence electrons.
(b) Use this list to construct the first two rows of the periodic table.
(c) Organic compounds often contain sulfur, phosphorus, chlorine, bromine, and iodine. Add these elements to your periodic table.

1-29 For each compound, state whether its bonding is covalent, ionic, or a mixture of covalent and ionic.
(a) KCl **(b)** KOH **(c)** CH_3CH_2Li **(d)** CH_3Cl
(e) $KOCH_3$ **(f)** CH_3CO_2Na **(g)** CCl_4

1-30 **(a)** Both PF_3 and PF_5 are stable compounds. Draw Lewis structures for these two compounds.
(b) NF_3 is a known compound, but all attempts to synthesize NF_5 have failed. Draw Lewis structures for NF_3 and a hypothetical NF_5, and explain why NF_5 is an unlikely structure.

1-31 Draw a Lewis structure for each species.
(a) N_2H_4 **(b)** N_2H_2 **(c)** $(CH_3)_2NH_2Cl$ **(d)** CH_3CN
(e) CH_3CHO **(f)** $CH_3S(O)CH_3$ **(g)** H_2SO_4 **(h)** CH_3NCO
(i) $CH_3OSO_2OCH_3$ **(j)** $CH_3C(NH)CH_3$ **(k)** $(CH_3)_3CNO$

1-32 Draw a Lewis structure for each compound. Include all nonbonding pairs of electrons.
(a) $CH_3COCH_2CHCHCOOH$ **(b)** $NCCH_2COCH_2CHO$
(c) $CH_2CHCH(OH)CH_2CO_2H$ **(d)** $CH_2CHC(CH_3)CHCOOCH_3$

1-33 Draw a line–angle formula for each compound in Problem 1-32.

1-34 Draw Lewis structures for
(a) two compounds of formula C_4H_{10} **(b)** two compounds of formula C_2H_6O
(c) two compounds of formula C_2H_7N **(d)** three compounds of formula C_2H_7NO
(e) three compounds of formula $C_3H_8O_2$ **(f)** three compounds of formula C_2H_4O

1-35 Draw a complete structural formula and a condensed structural formula for
(a) three compounds of formula C_3H_8O **(b)** five compounds of formula C_3H_6O

1-36 Some of the following molecular formulas correspond to stable compounds. When possible, draw a stable structure for each formula.

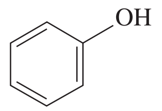
CH_2 CH_3 CH_4 CH_5

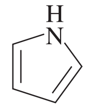
C_2H_2 C_2H_3 C_2H_4 C_2H_5 C_2H_6 C_2H_7

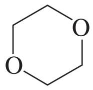
C_3H_3 C_3H_4 C_3H_5 C_3H_6 C_3H_7 C_3H_8 C_3H_9

Propose a general rule for the numbers of hydrogen atoms in stable hydrocarbons.

1-37 Draw complete Lewis structures, including lone pairs, for the following compounds.

(a) 
phenol

(b) 
pyrrole

(c) 
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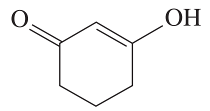
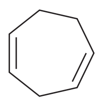
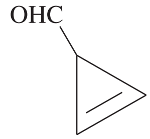
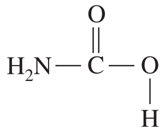
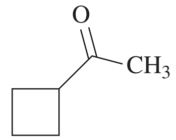
(d) 

(e) 

(f) 

(g) 

(h) 

1-38 Give the molecular formula of each compound shown in Problem 1-37.

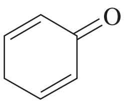
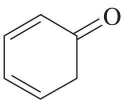
1-39 **1.** From what you remember of electronegativities, show the direction of the dipole moments of the following bonds.
2. In each case, predict whether the dipole moment is relatively large (electronegativity difference >0.5) or small.

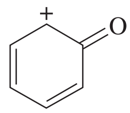
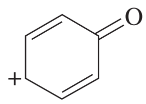
(a) C—Cl **(b)** C—H **(c)** C—Li **(d)** C—N **(e)** C—O
(f) C—B **(g)** C—Mg **(h)** N—H **(i)** O—H **(j)** C—Br

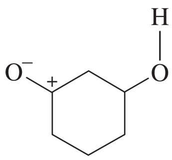
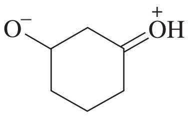
1-40 For each of the following structures,
1. Draw a Lewis structure; fill in any nonbonding electrons.
2. Calculate the formal charge on each atom other than hydrogen.

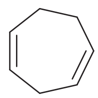
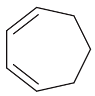
(a) CH_3NO (nitrosomethane) **(b)** $(CH_3)_3NO$ (trimethylamine oxide) **(c)** $[N_3]^-$ (azide ion)
(d) $[(CH_3)_3O]^+$ **(e)** CH_3NC **(f)** $(CH_3)_4NBr$

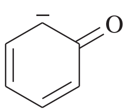
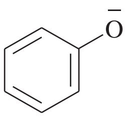
1-41 Determine whether the following pairs of structures are actually different compounds or simply resonance forms of the same compounds.

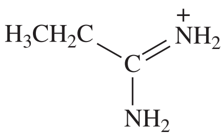
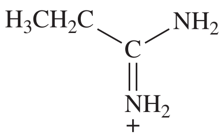
(a)  and 

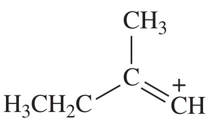
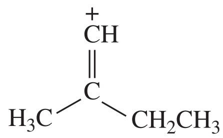
(b)  and 

(c)  and 

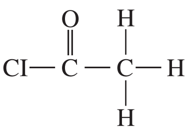
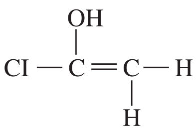
(d)  and 

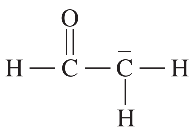
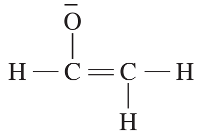
(e)  and 

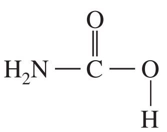
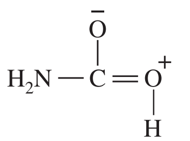
(f)  and 

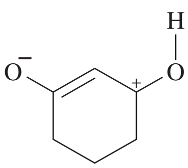
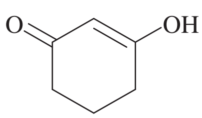
(g)  and 

(h) $CH_3CH_2CH{=}CHCH_3$ and $(CH_3)_2C{=}CHCH_3$

(i)  and 

(j)  and 

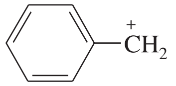
(k)  and 

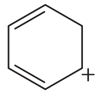
(l)  and 

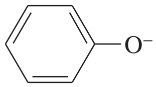
1-42 Draw the important resonance forms to show the delocalization of charges in the following ions. In each case, indicate the major resonance form(s).

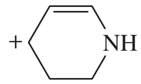
(a) $CH_3-\overset{\overset{O}{\|}}{C}-\overset{-}{C}H_2$

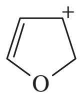
(b) $H-\overset{\overset{O}{\|}}{C}-CH{=}CH-\overset{-}{C}H_2$

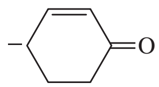
(c) 

(d) 

(e) 

(f) 

(g) 

(h) 

(i) $CH_2{=}CH-CH{=}CH-\overset{+}{C}H-CH_3$

(j) $CH_3-CH{=}CH-CH{=}CH-CH_2-\overset{+}{C}H_2$

(k) $\overset{+}{C}H_2-CH{=}CH-O-CH-CH_2$

1-43 In the following sets of resonance forms, label the major and minor contributors and state which structures would be of equal energy. Add any missing important resonance forms.

(a) $\left[CH_3-\underset{\cdot\cdot}{\overset{-}{C}}H-C\equiv N{:} \longleftrightarrow CH_3-CH{=}C{=}\ddot{N}{:}^{-}\right]$

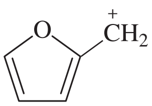
(b) $\left[CH_3-\overset{\overset{O^-}{|}}{C}{=}CH-\overset{+}{C}H-CH_3 \longleftrightarrow CH_3-\underset{+}{\overset{\overset{O^-}{|}}{C}}-CH{=}CH-CH_3\right]$

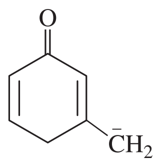
(c) $\left[CH_3-\overset{\overset{O}{\|}}{C}-\overset{-}{C}H-\overset{\overset{O}{\|}}{C}-CH_3 \longleftrightarrow CH_3-\overset{\overset{O^-}{|}}{C}{=}CH-\overset{\overset{O}{\|}}{C}-CH_3\right]$

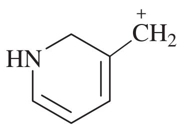
(d) $\left[CH_3-\overset{-}{C}H-CH{=}CH-NO_2 \longleftrightarrow CH_3-CH{=}CH-\overset{-}{C}H-NO_2\right]$

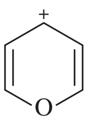
(e) $\left[CH_3-CH_2-\underset{+}{\overset{\overset{NH_2}{|}}{C}}-NH_2 \longleftrightarrow CH_3-CH_2-\overset{\overset{NH_2}{|}}{C}{=}\overset{+}{N}H_2\right]$

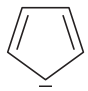
1-44 For each of these ions, draw the important resonance forms and predict which resonance form is likely to be the major contributor.

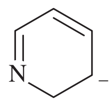
(a) 

(b) 

(c) 

(d) 

(e) 

(f) 

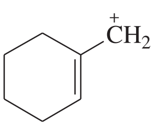
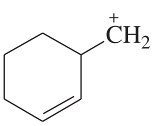
1-45 For each pair of ions, determine which ion is more stable. Use resonance forms to explain your answers.

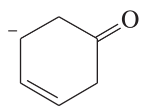
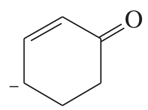
(a) $CH_3-\overset{+}{C}H-CH_3$ or $CH_3-\overset{+}{C}H-OCH_3$

(b) $\begin{array}{c} CH_3-N-CH_3 \\ | \\ CH_3-\underset{+}{C}-CH_3 \end{array}$ or $\begin{array}{c} CH_3-CH-CH_3 \\ | \\ CH_3-\underset{+}{C}-CH_3 \end{array}$

(c) $CH_2{=}CH-\overset{+}{C}H-CH_3$ or $CH_2{=}CH-CH_2-\overset{+}{C}H_2$

(d) $\overset{-}{C}H_2-CH_3$ or $\overset{-}{C}H_2-C\equiv N{:}$

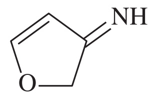
(e)  or 

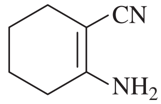
(f)  or 

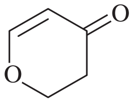
1-46 Use resonance structures to identify the areas of high and low electron density in the following compounds:

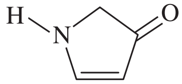
(a) $CH_3-\overset{\overset{\displaystyle O}{\|}}{C}-H$ **(b)** $CH_3-\overset{\overset{\displaystyle O}{\|}}{C}-NH_2$ **(c)** $CH_3-\overset{\overset{\displaystyle NH}{\|}}{C}-H$ **(d)** $CH_3-\overset{\overset{\displaystyle O}{\|}}{C}-OCH_3$

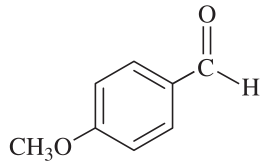
(e) $CH_3O-\overset{\overset{\displaystyle O}{\|}}{C}-NH_2$

(f) 

(g) 

(h) 

(i) 

(j) 

1-47 Compound X, isolated from lanolin (sheep's wool fat), has the pungent aroma of dirty sweatsocks. A careful analysis showed that compound X contains 62.0% carbon and 10.4% hydrogen. No nitrogen or halogen was found.
(a) Compute an empirical formula for compound X.
(b) A molecular weight determination showed that compound X has a molecular weight of approximately 117. Find the molecular formula of compound X.
(c) Many possible structures have this molecular formula. Draw complete structural formulas for four of them.

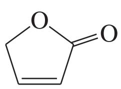
*__1-48__ In 1934, Edward A. Doisy of Washington University extracted 3000 lb of hog ovaries to isolate a few milligrams of pure estradiol, a potent female hormone. Doisy burned 5.00 mg of this precious sample in oxygen and found that 14.54 mg of CO_2 and 3.97 mg of H_2O were generated.
(a) Determine the empirical formula of estradiol.
(b) The molecular weight of estradiol was later determined to be 272. Determine the molecular formula of estradiol.

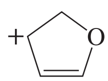
1-49 If the carbon atom in CH_2ClBr were flat, there would be two stereoisomers. The carbon atom in CH_2ClBr is actually tetrahedral. Make a model of this compound, and determine whether there are any stereoisomers of CH_2ClBr.

$$\begin{array}{ccc} & H & \\ & | & \\ H- & C & -Cl \\ & | & \\ & Br & \end{array} \qquad \begin{array}{ccc} & H & \\ & | & \\ Br- & C & -Cl \\ & | & \\ & H & \end{array}$$

1-50 Cyclopropene (C_3H_4, a three-membered ring) is more reactive than most other cycloalkenes.
(a) Draw a Lewis structure for cyclopropene.
(b) Compare the bond angles of the carbon atoms in cyclopropene with those in an acyclic (noncyclic) alkene.
(c) Suggest why cyclopropene is so reactive.

1-51 For each of the following compounds,
1. Give the hybridization and approximate bond angles around each atom except hydrogen.
2. Draw a three-dimensional diagram, including any lone pairs of electrons.
(a) H_3O^+ **(b)** ^-OH **(c)** CH_2CHCN **(d)** $(CH_3)_3N$ **(e)** $[CH_3NH_3]^+$
(f) CH_3COOH **(g)** CH_3CHNH **(h)** CH_3OH **(i)** CH_2O

1-52 For each of the following compounds and ions,
1. Draw a Lewis structure.
2. Show the kinds of orbitals that overlap to form each bond.
3. Give approximate bond angles around each atom except hydrogen.
(a) $[NH_2]^-$ **(b)** $[CH_2OH]^+$ **(c)** $CH_2=N-CH_3$
(d) $CH_3-CH=CH_2$ **(e)** $HC\equiv C-CHO$ **(f)** H_2N-CH_2-CN
(g) $CH_3-\overset{\overset{\displaystyle O}{\|}}{C}-OH$ **(h)** 

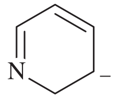
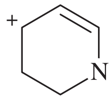
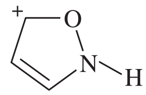
(i) 

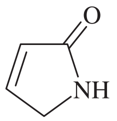
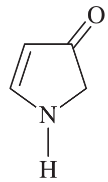
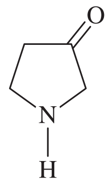
1-53 In most amines, the nitrogen atom is sp^3 hybridized, with a pyramidal structure and bond angles close to 109°. In urea, both nitrogen atoms are found to be planar, with bond angles close to 120°. Explain this surprising finding. (*Hint*: Consider resonance forms and the overlap needed in them.)

$$H_2\ddot{N}-\overset{\overset{\displaystyle O}{\|}}{C}-\ddot{N}H_2$$
urea

1-54 Predict the hybridization and geometry of the carbon and nitrogen atoms in the following molecules and ions. (*Hint:* Resonance.)

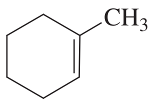
(a) $CH_3-\overset{\overset{\displaystyle O}{\|}}{C}-\overset{-}{C}H_2$ **(b)** $H_2N-CH=CH-\overset{+}{C}H_2$ **(c)** $\overset{-}{C}H_2-C\equiv N$

(d)  **(e)**  **(f)** 

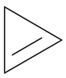
(g)  **(h)**  **(i)** 

1-55 Draw orbital pictures of the pi bonding in the following compounds:
(a) CH_3COCH_3 **(b)** HCN **(c)** $CH_2=CH-CHCHCN$
(d) $CH_3C\equiv CCHO$ **(e)** $CH_3CH=C=CHCH_3$ **(f)** $CH_3-CH=N-CH=C=O$

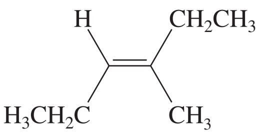
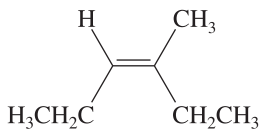
1-56 **(a)** Draw the structure of *cis*- $CH_3-CH=CH-CH_2CH_3$ showing the pi bond with its proper geometry.
(b) Circle the six coplanar atoms in this compound.
(c) Draw the trans isomer, and circle the coplanar atoms. Are there still six?
(d) Circle the coplanar atoms in the following structure:

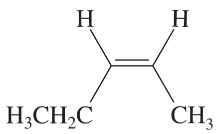
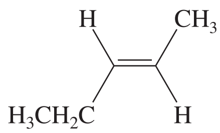


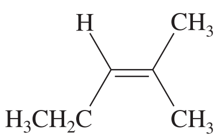
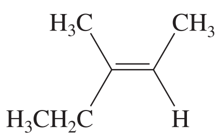
1-57 In pent-2-yne ($CH_3CCCH_2CH_3$), there are four atoms in a straight line. Use dashed lines and wedges to draw a three-dimensional representation of this molecule, and circle the four atoms that are in a straight line.

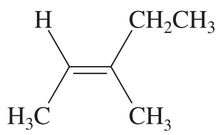
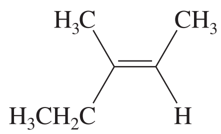
1-58 Which of the following compounds show cis-trans isomerism? Draw the cis and trans isomers of the ones that do.
(a) $CH_3CH_2CH=CHCH_3$ **(b)** $HC\equiv CH$ **(c)** $(CH_3)_2C=CHCH_3$
(d) cyclopropene,  **(e)** $CH_3CH_2CH=\underset{\underset{\displaystyle CH_2CH_3}{|}}{C}CH_3$ **(f)** $CH_3CH_2CH=\underset{\underset{\displaystyle H}{|}}{N}$

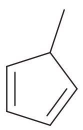
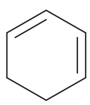
1-59 Give the relationships between the following pairs of structures. The possible relationships are as follows: same compound, cis-trans isomers, constitutional (structural) isomers, and not isomers (different molecular formula).
(a) $CH_3CHCHCH_3$ and $(CH_3)_3CH$ **(b)** CH_2CClCH_3 and CH_2CHCH_2Cl

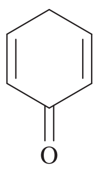
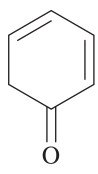
(c)  and 

(d)  and 

(e)  and 

(f)  and 

(g)  and 

(h)  and 

***1-60** Dimethyl sulfoxide (DMSO) has been used as an anti-inflammatory rub for race horses. DMSO and acetone appear to have similar structures, but the C=O carbon atom in acetone is planar, while the S=O sulfur atom in DMSO is pyramidal. Draw Lewis structures for DMSO and acetone, predict the hybridizations, and explain these observations.

$CH_3-\overset{\overset{\displaystyle O}{\|}}{S}-CH_3$ $CH_3-\overset{\overset{\displaystyle O}{\|}}{C}-CH_3$

DMSO acetone